Labor rigidities meet financial constraints: Evidence from Italy

Erin P. Gibson*
University of Michigan

November 9, 2026

Updated frequently. Click here for most recent version.

Abstract

This paper explores how rising unit labor costs, tied to nationally set contractual wage floors, depressed the Italian manufacturing sector. Using firm-level panel data, I document a negative relationship between sectoral wage floor growth and firm-level investment, employment, and sales. This relationship is highly heterogeneous and suggests a role for financial frictions: Small, highly leveraged firms experience contractions three times larger than those of large, low-leverage firms. To understand the mechanisms behind this and quantify aggregate effects, I develop a model where overlapping generations of heterogeneous firms face both collateral constraints and industry-wide wage floors. Calibrated to the 20 percent increase in unit labor costs incurred by the Italian manufacturing sector from 1999 to 2015, the model predicts aggregate contractions in manufacturing investment (4.8 percent), employment (6.1 percent), and real GDP (5.3 percent). A decomposition reveals that financial frictions amplify the direct effects of wage rigidity by approximately 30 percent through an interaction effect. The results highlight how labor market institutions and financial frictions can interact to generate persistent effects on firm dynamics and aggregate performance.

*I am deeply grateful to my committee, Linda Tesar, Chris House, Pablo Ottonello, and Kathryn Dominguez, for their comments and insights throughout this project.

1 Introduction

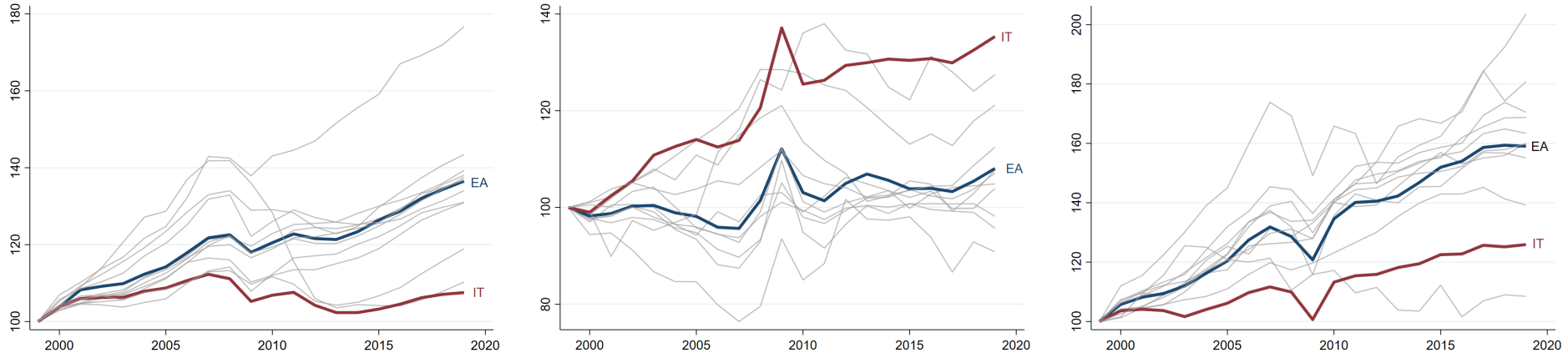
Despite being the euro area's third-largest economy, Italy has remained largely stagnant since adopting the euro in 1999. Panel (a) of Figure 1 plots real GDP indexed to 1999 for Italy alongside the euro area average. As of 2019, Italy's real GDP was only about 7 percent above its 1999 level, while the euro area average grew by nearly 40 percent over the same period. This sluggish overall growth cannot be reduced to a consequence of the financial or euro debt crises: Even prior to 2007, Italy's GDP growth lagged well below the euro area average.

Over the same period, Italian real wage growth outpaced gains in labor productivity. As shown in panels (b) and (c) of Figure 1, real unit labor costs (ULCs) in the Italian manufacturing sector rose by approximately 25 percent between 1999 and 2015—far exceeding the modest productivity gains achieved over the same period and significantly outpacing other euro area countries. While ULC growth stalled following the global financial crisis, the level remained elevated well above the euro area average, and labor productivity growth has remained low. This trend coincided with sharp declines in net investment and employment in the manufacturing sector, as illustrated in Figure 2.

This paper examines how ULC growth has undermined growth in the Italian manufacturing sector over the past two decades. The analysis centers on Italy's system of centralized collective bargaining, which determines sector- and occupation-specific wage floors at the national level. These wage floors are binding for the majority of firms and thus provide an effective proxy for ULCs. To understand aggregate stagnation, I consider firm-level investment and employment responses to rising labor costs. While labor allocations respond directly to higher wage floors, two economic mechanisms can rationalize a strong negative response of investment to ULC growth. First, due to factor complementarity, cost-induced reductions in labor lower the marginal product of capital and deter investment. Second, higher wage bills and lower production can erode a firm's net worth and restrict its ability to borrow when subject to collateral constraints. Without sufficient access to funds, firms invest below their optimal levels, further hindering growth. Thus, in studying the effects of ULC growth, I also examine the interaction between wage rigidities and financial frictions. Specifically, I consider two main questions: How does ULC growth affect firm-level and aggregate outcomes? And how do financial frictions amplify these effects? This paper provides both empirical and theoretical answers using firm-level data on Italian manufacturing firms from 1999-2019 and a small open economy model featuring heterogeneous firms, financial frictions, and wage floors.

The empirical analysis combines sector-specific wage floor data with firm-level balance sheet data to explore the relationship between wage floor growth and firm-level

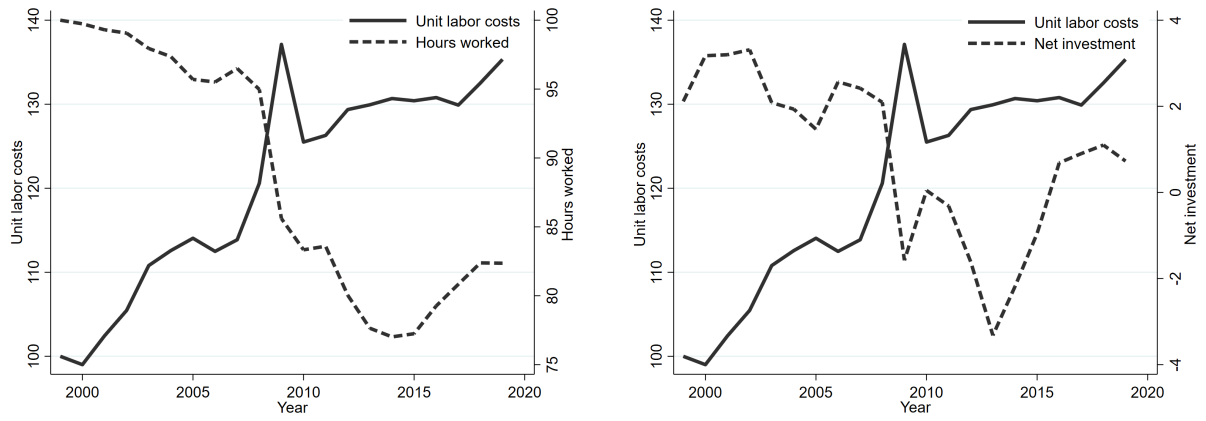
Figure 1: GDP, unit labor cost, and labor productivity in Italy and the euro area



(a) GDP (b) Unit labor cost (c) Labor productivity

Note: All variables expressed in real values and indexed to 1999=100. The red line corresponds to Italy, the blue line to the euro area average, and the unlabeled gray lines correspond to the remaining 12 original euro area countries. ULC is the ratio of employee compensation to real value added. Labor productivity is real output per hour. ULC and productivity are for manufacturing. Source: Eurostat.

Figure 2: Unit labor cost, hours worked, and net investment



(a) Hours worked

(b) Net investment

Note: Unit labor cost (ULC) is defined as the ratio of compensation of employees to real value added. Hours worked, and ULC are indexed to 1999 = 100. Net investment is reported as a percentage of value added. Source: Eurostat.

outcomes. By interacting sectoral wage floor growth with firm-level labor intensity, I proxy changes in firm-level labor costs. The empirical analysis yields two facts: First, increases in contractual wage floors are associated with contractions in firm-level investment, employment, and sales. A one percentage point increase in firm-level labor costs (the interaction of sectoral wage floor growth and the firm's lagged labor share) is correlated with a 0.40 percentage point contraction in investment, a 0.32 percentage point contraction in employment, and a 0.28 percentage point contraction in sales. Second, these effects are strongly heterogeneous across firms: Sectoral wage floor growth correlates with much larger contractions for small and high-leverage firms than for large and low-leverage firms, consistent with a potential role for financial frictions in amplifying the effects of wage floor growth.

To understand the mechanisms behind these heterogeneous relationships, I develop and calibrate a small open economy model that features overlapping generations of entrepreneurs who face both labor market and financial frictions. In the model, entrepreneurs produce differentiated intermediate goods and hire workers in monopsonistic markets subject to industry-wide wage floors. Entrepreneurs can access financing from international bond markets but are potentially limited by the presence of a collateral constraint tied to their net worth. The model generates predictions consistent with the empirical patterns and allows for quantification of the aggregate and welfare costs of the 20 percent wage floor increase. Additionally, the model enables isolation of the contributions of wage floor growth, financial frictions, and their interaction using counterfactual analyses.[1]

[1]Mechanically, there are two ways in which the wage floor and collateral constraint interact in the model outside of general equilibrium effects, a "static" interaction and a "dynamic" interaction. For the

The quantitative analysis reveals that the 20 percent increase in contractual wage floors reduced aggregate manufacturing investment by 4.8 percent, employment by 6.1 percent, and real GDP by 5.3 percent. Consistent with the empirical patterns, small and highly leveraged firms experienced the largest contractions, while some large firms actually expanded by gaining market share from constrained competitors. The welfare effects of the increase are negative, with both workers and entrepreneurs suffering consumption losses. Financial frictions magnify the damage from the wage floor increase: The decomposition reveals a roughly 30 percent amplification effect on investment through the interaction between collateral constraints and higher wage floors. Additional counterfactual exercises reveal significant gains to both financial and labor market reforms, as well as a role for openness in worsening the impact of high wage floors.

The findings have important implications for understanding Italian economic stagnation and for designing labor market reforms. While wage moderation is crucial, measures to reduce financial frictions and improve access to financing are equally important in restoring Italy's competitiveness. More broadly, the analysis highlights how interactions between different market imperfections can generate outcomes that are more severe than the sum of their individual effects.

Literature review This paper relates to four main strands of literature. First, it contributes to the macroeconomic discussion of Italy's stagnation and struggles since joining the euro area. Several theories have been proposed to explain this phenomenon: One version focuses on slow structural reform, citing strict employment protection, strong trade unions, an expensive welfare state, overgenerous pensions, and anti-competitive product market regulation (e.g., Daveri et al. 2000; Alesina et al. 2008; Tokarski 2019). A complementary view focuses on how differences in wage bargaining systems led to higher inflation in Italy than in other euro area states. Due to the fixed exchange rate, this hurt the competitiveness of Italian firms and depressed long-run growth (e.g., Hancké 2013; Hall 2014; Bofinger 2015; Kangur 2018). Additionally, several papers detail firm-level characteristics and important drivers of low productivity growth. In particular, the small size of Italian firms has been widely studied (e.g., Amatori et al. 2013; Bugamelli et al. 2018), as has their reliance on family management (Bandiera et al. 2015; Pellegrino and Zingales 2017). I connect the macro narratives around the costs of national bargaining to firm-level behavior, mapping contractual wage floor growth into firm-level adjustment and aggregate outcomes. Additionally, my framework offers an explanation for the small

"static" interaction, a financially constrained firm is more constrained under any given wage floor than a non-financially constrained firm with the same productivity, because its lower capital level leads to a lower marginal revenue product of labor. For the "dynamic" interaction, at higher levels of the wage floor, firms accumulate net worth more slowly and thus remain financially constrained (and, consequently, stuck at low levels of investment) for longer than they otherwise would.

size of Italian firms as restrictively high wage floors suppress growth.

Second, this paper relates to the recent literature on modeling the general equilibrium effects of a minimum wage. Engbom and Moser (2022), Hurst et al. (2022), and Drechsel-Grau (2023) propose search models to study the distributional impact of the minimum wage across heterogeneous households. Ahlfeldt et al. (2022) and Karabarbounis et al. (2022) focus on the spatial dimensions and the effect on entry decisions, respectively. Closest to this paper, Berger et al. (2025) study the effects of minimum wages in the monopsonistic competition model of Berger et al. (2022). My model treats Italy's sectoral wage floors as a manufacturing minimum wage and builds on Berger et al. (2025)'s framework by incorporating financial frictions and open economy dimensions–features that have been largely underexplored in the minimum wage literature.

Additionally, I connect to a large literature on the effects of financial frictions. A dominant strand of this literature looks at the effects of financial frictions on resource allocation and productivity through heterogeneous firm models (e.g., Hopenhayn and Rogerson 1993; Buera et al. 2011; Midrigan and Xu 2014). Most relevantly, several studies examine the impact of financial frictions in open economy settings: Gopinath et al. (2017) and Asriyan et al. (2022) study the impact of financial integration, modeled as a fall in the world real interest rate, on open economies with heterogeneous firms and collateral constraints. Andreasen et al. (2023) and Andreasen et al. (2024) consider the impact of capital controls in an open economy model with financial frictions. I expand on this literature by incorporating labor market frictions (in the form of sectoral wage floors) into an open economy model with heterogeneous firm collateral constraints. In this environment, I can quantify how these market imperfections interact to affect firm-level and aggregate outcomes.

Finally, this paper contributes to a small literature of empirical studies on how labor market imperfections influence investment and capital-labor substitution decisions. This has been more commonly studied through the lens of employment protection legislation (EPL). For example, Calcagnini et al. 2009 and Cingano et al. 2010 conduct panel studies of the effects of EPL on European firms, and Autor et al. 2007 studies the effects in the United States. Most relevant to this paper, Garcia-Macia (2020) estimates the effect of changes in Italian wage floors on profitability and investment. This literature has primarily focused on estimating elasticities, thus leaving room for a deeper understanding of underlying economic mechanisms. I contribute by developing a model to understand how wage floor increases interact with financial frictions to affect investment, as well as other firm-level and aggregate outcomes.

The remainder of the paper is organized as follows: Section 2 reviews the institutional background, describes the empirical methodology, and presents the main firm-level results.

Section 3 develops the theoretical model. Section 4 characterizes the equilibrium and discusses the mechanisms underlying firm responses to wage floors. Section 5 presents the model calibration and quantitative results. Section 6 concludes with policy implications and directions for future research.

2 Empirical analysis

This section explores the relationship between unit labor cost growth and firm-level outcomes in the Italian manufacturing sector. To do this, I exploit Italy's institution of nationalized wage bargaining. Combining nationally set sectoral wage floors with firm-level data, I explore how firms adjust capital, labor, and sales in response to changing labor costs. Documenting heterogeneity across firms of different size and leverage classes provides some initial descriptive evidence for the role of financial frictions in amplifying the effects of wage floor growth.

2.1 Institutional context

Rising unit labor costs in Italy are closely linked to the country's collective bargaining system. Contracts setting sector- and occupation-specific wage floors are negotiated at the national level between trade unions and employers' associations, and their provisions are extended to nearly all firms and workers. These contracts have virtually universal coverage even for firms and employees not represented in the bargaining process because labor courts generally use negotiated wage floors as a reference for a fair wage.[2] As a result, contracts cover the vast majority of the formal economy: There are 346 national agreements covering 97.7 percent of dependent workers and 99.3 percent of firms, and non-compliance with contractual wages in Italy is extremely rare (Adamopoulou and Villanueva 2022).

The Italian system does permit limited firm-level bargaining. High-productivity firms can pay firm-level bonuses above the negotiated industry floors. Additionally, limited reforms have increased the prevalence of fixed-term contracts, allowing firms to pay wages below the contractual minima to a limited share of workers (Daruich et al. 2023). However, the negotiated minima are effectively binding.[3] Survey evidence in D'Amuri and

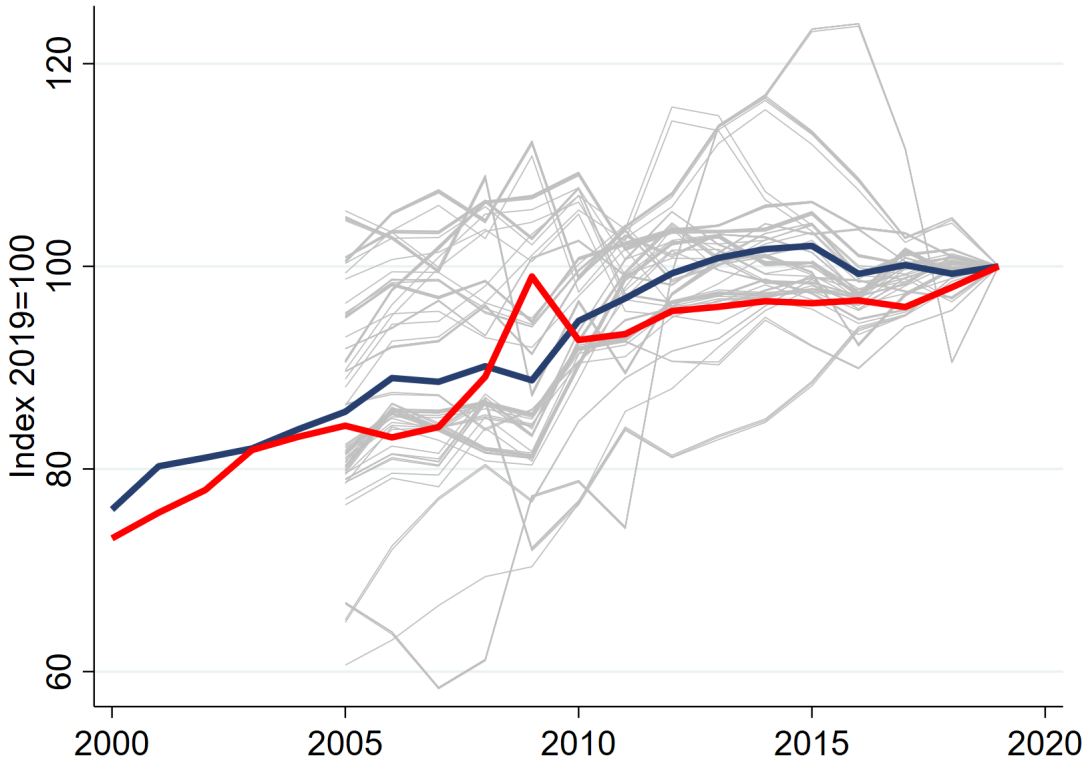
[2]The extension of contractual wages to firms and workers outside of negotiations has been shown to have economic consequences: Extensions negatively affect employment growth (Martins 2021; Hijzen and Martins 2016) and can be used as a tool for "insider" firms to drive competitors out of the market (Haucap et al. 2001; Magruder 2012).

[3]Boeri et al. (2021) show that these minima are especially binding in Southern Italy, resulting in disproportionately high unemployment.

Giorgiantonio (2015) reveals that only around 10.5 percent of wages exceed the minima set by the national collective agreements.[4]

Several studies have documented the economic consequences of this bargaining structure. Prior work highlights the contribution of collective bargaining to wage inequality (Manacorda 2004; Devicienti et al. 2019; Leonardi et al. 2019), downward wage rigidity (Devicienti et al. 2007; Diéz-Catalán and Villanueva 2015), regional differences in employment (Boeri et al. 2021), and exchange rate overvaluation (Kangur 2018).

Figure 3: Sectoral wage floors and unit labor costs



The red line plots unit labor cost (ULC) for the Italian manufacturing sector, defined as the ratio of compensation of employees to real GDP. The blue line plots the index of real wages according to collective labor agreements for the manufacturing sector as a whole. The gray lines plot the same series for all manufacturing sub-sectors at the NACE 3-digit level. Wages are deflated using NACE 2-digit value added deflators. All series are normalized such that 2019 = 100. Source: IStat, Eurostat.

Given their binding nature, growth in negotiated sectoral wage floors provides a sufficient proxy for ULC growth in Italy's manufacturing sector. Figure 3 illustrates the close correspondence between manufacturing ULCs (in red) and the aggregate contractual wage index for the manufacturing sector (in blue). Both measures rose by about 20 percent between 2000 and 2015 before stabilizing at an elevated level.

[4]This finding is consistent with other studies: A 1995 survey of 8,000 firms in the manufacturing and service sectors found that only 10 percent of firms reported engaging in firm-level bargaining (IStat 2000). Further studies have shown that this share has declined as wage floor increases accelerated following euro area integration (see e.g., Destefanis et al. 2005 and Brandolini et al. 2007).

2.2 Data

The empirical analysis combines firm-level balance sheet data with sector-level data on nationally set contractual wage floors. The close link between negotiated minima and ULCs documented in Figure 3 supports the use of these contractual floors in measuring changing labor costs.

Firm level data: I construct a panel dataset of Italian manufacturing firms using the Orbis global database, following the cleaning and harmonization procedures in Kalemli-Özcan et al. (2024). The sample spans 1999–2019 and includes detailed balance sheet information, including assets, value added, investment, profits, debt, and employment, allowing for the computation of firm-level labor shares and productivity. After applying the vintaging adjustments, the data cover between 70 and 90 percent of Italian manufacturing output across the sample years and are broadly representative of firms across the size distribution.[5] All nominal variables are converted into real terms using the NACE-2 deflators for value added and investment from Eurostat. Table A.1 provides summary statistics for key variables.

Sectoral wage floors: The Italian collective bargaining system is highly centralized. Collective agreements are signed at the national level for each industry, with little room for firm-level adjustment. Contracts assign wage floors for workers across different categories and professional classes. While I do not observe the full set of detailed contract provisions, I use IStat's publicly available indices of average contractual wages at the NACE 3-digit level for 2005-2019.[6] For the manufacturing sector, this yields roughly 120 different sub-industries. These indices are weighted averages of the contractual minima set by the underlying agreements, capturing both renewals and the staggered implementation of previously agreed pay increases.

The gray lines in Figure 3 show the real contractual wage floor indices at the NACE-3 level and reveal substantial heterogeneity across sub-sectors. Annual changes in contractual wage floors range from –3 to +8 percent across industries and years, providing substantial variation to exploit in the regression analysis.

[5] This is validated in Tables A.2 and A.3.

[6] Devicienti and Fanfani (2025) show that contractual wage growth is highly correlated within an industry across worker classes. Given this and my lack of employer-employee linked data, I apply the sectoral average across all occupations and assume all classes of workers are equally affected.

2.3 Empirical strategy

The empirical model aims to estimate the relationship between changes in nationally set sectoral wage floors and firm-level outcomes. Following Garcia-Macia (2020), the identifying variation comes from the interaction between sector-level contractual wage floor growth and firm-level exposure to wage changes, captured by the lagged labor share. All firms within a sector are subject to the same contractual floors, but vary in their labor intensity. Thus, changes in the contractual wage floor translate into heterogeneous cost changes across firms. The baseline specification is

$$\%\Delta Y_{ijt} = \omega_0 + \omega_1 \left(\alpha_{ijt-1} \times \%\Delta \bar{w}_{jt}\right) + \omega_3 X_{ijt-1} + \delta_i + \delta_{jt} + \varepsilon_{it}, \tag{1}$$

where the subscripts refer to firm i in NACE-3 industry j at time t. $\%\Delta Y_{ijt}$ represents the percent change in firm-level outcome variables of interest: Capital, employment, and sales. The main explanatory variable is the firm's lagged labor share, α_{ijt-1}, interacted with the rate of growth of the contractual wage floor in its industry, $\%\Delta \bar{w}_{jt}$. The vector X_{ijt-1} includes standard firm-level controls, such as the lagged dependent variable, the lagged labor share, leverage, debt service ratio, cash flow ratio, and size tercile.

Firm fixed effects, δ_i, control for time-invariant heterogeneity across firms, and sector-year fixed effects, δ_{jt}, absorb aggregate shocks common to all firms within a sector. Standard errors are clustered at the sector-year level. Thus, coefficients are identified from within-firm variation in exposure over time relative to other firms in the same industry-year. I interpret the estimates as conditional correlations between changes in contractual wage floors and firm-level adjustments.

2.4 Empirical results

Table 1 reports the results of the main regression specification for capital, labor, and sales growth, restricting to the coefficient of interest on the interaction between sectoral wage floor growth and firm-level lagged labor share. Full results are reported in Table A.4.

For all three dependent variables, the estimates for ω_2 are negative and statistically significant, confirming a negative relationship between contractual wage-floor growth and firm-level capital, employment, and sales growth. A one percentage point increase in firm-level labor costs (measured by the interaction of lagged labor share with contractual wage floor growth) is associated with a 0.40 percentage point contraction in investment, a 0.32 percentage point contraction in employment, and a 0.28 percentage point contraction in sales. Interestingly, the estimated coefficient is largest for capital growth, suggesting a potential effect beyond factor complementarity. To explore the mechanisms behind

this, I consider a potential role for financial frictions: in an environment with collateral constraints, higher wage bills and lower production can erode a firm's net worth and constrain its borrowing capacity, leading to lower investment.

Table 1: Contractual wage floor growth and firm outcomes

	Capital Growth	Labor Growth	Sales Growth
$\alpha_{ijt-1} \times \%\Delta\bar{w}_{jt}$	-0.399^{**}	-0.320^{***}	-0.275^{**}
	(0.126)	(0.163)	(0.140)
Observations	405,092	339,666	405,061
Adjusted R^2	0.360	0.123	0.451
Firm-FE	YES	YES	YES
Sector-year FE	YES	YES	YES

Notes: This table reports the main estimated coefficient from Equation 1. The sample period is 2005-2019. Standard errors in parentheses clustered at the sector-year level. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

I provide initial evidence for this hypothesis by interacting the main independent variable, $\alpha_{ijt-1} \times \%\Delta\bar{w}_{jt}$, with indicators of firm size and leverage, where size is measured as the book value of assets. In an environment with collateral constraints, small and highly leveraged firms are the most likely to be constrained in borrowing, while large and low-leverage firms tend to avoid such constraints. Additionally, any existing financial constraints likely tightened during the financial crisis. To capture this, I also conduct sub-period analyses and report separate estimates for crisis (2009 to 2014) and non-crisis years. Thus, if financial constraints amplify the effects of wage floor growth, the estimated negative relationship should be larger for small or highly levered firms and for all firms during the crisis period, relative to large or low-leverage firms and during the non-crisis period.

Table 2 reports the estimates for the main coefficient in each of these regressions. Looking first at the estimates by tercile of leverage, the negative relationship is most pronounced for highly leveraged (top tercile) firms. Comparing the top and bottom terciles, highly levered firms contract investment by roughly 2.5 times and employment by roughly 2.1 times the amount that low-leverage firms do in response to a 1 percentage point increase in labor costs. Comparing firms by size tercile, small (bottom tercile) firms show an especially strong negative relationship for each dependent variable. By contrast, large (top tercile) firms actually see positive, albeit largely insignificant, estimates, suggesting that the largest first may gain market share as the smallest shrink in response to wage floor growth. Interacting size and leverage reinforces these results: Small and highly levered firms exhibit the strongest contractions, followed by small and low-leverage firms. Large firms show no significant relationship for either leverage class, consistent with the idea that

firms can outgrow collateral constraints; once a firm is large enough, additional investment can be financed internally. The euro area debt crisis period (2009–2014) provides some additional support for the role of financial frictions in amplifying the effects of wage floor growth. The negative estimates are most pronounced during the crisis period, when financing constraints were most severe.

Robustness The results are stable across a range of alternative specifications. Robustness exercises include removing various firm-level controls, replacing sector-year fixed effects with year fixed effects, and extending the differences to two-year horizons. Although they vary in statistical significance, all alternative specifications yield negative estimated coefficients on capital, labor, and sales growth. Estimates are reported in the Appendix A.2.

Additionally, I explore a role for trade openness in amplifying the effects of rising labor costs. To do this, I interact the firm-level cost change, $\alpha_{ijt-1} \times \%\Delta\bar{w}_{jt}$, with sectoral exposure to Chinese imports. Exposure to imports should amplify the negative effects of increases in contractual wage floors because firms exposed to greater competitive pressures are less able to pass the burden of higher wages to consumers via higher prices. Sector j's exposure to Chinese import competition is defined as $CE_j = \text{IM}_{j,2001}^{China}/\text{VA}_{j,2001}$, Where $\text{IM}_{j,2001}^{China}$ represents the 2001 value of imports from China in sector j and $\text{VA}_{j,2001}$ represents sector j's 2001 total value added. To address endogeneity, CE_j is constructed using import and value added data for the United States. Thus, the measure of exposure reflects Chinese supply-side competitiveness rather than Italian demand. The regression specification compares firms in NACE–3 industries with high versus low exposure to Chinese import competition,

$$\begin{aligned}\%\Delta Y_{ijt} = \omega_0 + \omega_1 \left(\alpha_{ijt-1} \times \%\Delta\bar{w}_{jt} \times \text{CE}_j^{\text{low}}\right) + \omega_2 \left(\alpha_{ijt-1} \times \%\Delta\bar{w}_{jt} \times \text{CE}_j^{\text{high}}\right) \\ + \omega_3 \text{CE}_j^{\text{low}} + \omega_4 \text{CE}_j^{\text{high}} + \omega_5 X_{ijt-1} + \delta_i + \delta_{jt} + \varepsilon_{it}.\end{aligned} \tag{2}$$

Here, $\text{CE}_j^{\text{high}}$ and CE_j^{low} are indicators representing whether sector j's exposure to Chinese import competition is above or below the median.

Table 3 reports the results of the regression specification in Equation 2 for capital, labor, and sales growth. The results strongly support the competition-amplification hypothesis. The estimated coefficients are much larger in magnitude for high-import-exposure industries than for low-import-exposure industries across all dependent variables, suggesting that increases in sectoral wage floors are associated with larger contractions in investment,

Table 2: Wage floor growth and firm outcomes by size, leverage, and period

Dependent variable	Capital Growth	Labor Growth	Sales Growth
Effects by period			
2009-2012	-0.415***	-0.489 **	-0.336*
	(0.169)	(0.218)	(0.198)
Excl. crisis	-0.375**	-0.142	-0.202*
	(0.199)	(0.214)	(0.118)
Effects by terciles of leverage			
Bottom tercile	-0.258*	-0.183	-0.218
	(0.133)	(0.167)	(0.146)
Middle tercile	-0.292**	-0.351**	-0.270*
	(0.135)	(0.166)	(0.141)
Top Tercile	-0.666***	-0.400**	-0.327**
	(0.134)	(0.170)	(0.150)
Effects by terciles of size			
Bottom tercile	-0.612***	-0.485***	-0.428***
	(0.143)	(0.145)	(0.141)
Middle tercile	-0.097	-0.381**	-0.018
	(0.129)	(0.177)	(0.145)
Top Tercile	0.252*	-0.058	0.280
	(0.140)	(0.156)	(0.158)
Effects by size and leverage			
Small and low leverage	-0.423**	-0.247*	-0.187*
	(0.192)	(0.148)	(0.113)
Large and low leverage	0.172	-0.084	0.187
	(0.132)	(0.165)	(0.151)
Small and high leverage	-0.616***	-0.587***	-0.433***
	(0.137)	(0.180)	(0.143)
Large and high leverage	-0.037	-0.110	0.136
	(0.141)	(0.164)	(0.165)

Notes: This table shows the main estimated coefficient on the interaction between wage floor growth and lagged labor share for different groups of firms. Size is defined as the book value of assets. Small, low-leverage firms are those with assets below the median and leverage above the median. Other categories are defined similarly. The sample period is 2005-2019. Standard errors in parentheses clustered at the sector-year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Exposure to China and firm response to sectoral wage growth

	Capital Growth	Labor Growth	Sales Growth
Low exposure to China (ω_1)			
$\alpha_{ijt-1} \times \%\Delta\bar{w}_{jt}$	-0.245^{**}	-0.198^{**}	-0.167^{*}
	(0.102)	(0.089)	(0.095)
High exposure to China (ω_2)			
$\alpha_{ijt-1} \times \%\Delta\bar{w}_{jt}$	-0.578^{***}	-0.467^{***}	-0.398^{***}
	(0.143)	(0.125)	(0.131)
Difference ($\omega_2 - \omega_1$)	-0.333^{***}	-0.269^{***}	-0.231^{**}
	(0.098)	(0.087)	(0.092)
Observations	375,673	329,527	375,456
Adjusted R^2	0.351	0.125	0.454
Firm-FE	YES	YES	YES
Sector-year FE	YES	YES	YES

Notes: This table reports the main estimated coefficients from Equation 2. The sample period is 2005-2019. Standard errors in parentheses clustered at the sector-year level. * ($p < 0.10$), ** ($p < 0.05$), *** ($p < 0.01$).

employment, and sales for firms in industries more exposed to Chinese import competition. The differential effect between categories is economically and statistically significant across all outcomes. Firms in industries facing high levels of Chinese import competition experience contractions in investment and employment that are roughly 2.4 times larger than those in more protected industries. This suggests that trade exposure significantly constrains firms' ability to adjust to rising labor costs, consistent with the hypothesis that competitive pressure limits the cost pass-through option.

Overall, the empirical evidence shows a robust negative relationship between growth in contractual wage floors and firm-level outcomes. This relationship is strongest for small and highly leveraged firms and for firms in sectors with strong import competition. However, the estimates are not causal. Even after controlling for firm and sector-year fixed effects and testing a variety of robustness checks, concerns remain about the endogeneity of contractual wage floors. This motivates further exploration of the economic channels underlying these results through the use of a model.

3 Model

Motivated by the empirical findings, I build and calibrate a general equilibrium model of firm dynamics with financial frictions, labor market power, and an economy-wide wage floor. My set-up incorporates elements from the open economy model of heterogeneous

firms and financial frictions used in Andreasen et al. (2023) and Andreasen et al. (2024) and from the model of monopsonistic firms and minimum wages[7] used in Berger et al. (2025). Using the model, I compare the equilibrium of a baseline specification, calibrated to the Italian manufacturing sector in 1999 (before the rise in labor costs), to the equilibrium once wage floors are increased by the 20 percent observed in the data. This enables me to study the aggregate and firm-level effects of the large increase in labor costs on Italian manufacturing firms and to explain the heterogeneous effects of increasing wage floors for firms of different size and leverage categories. Additionally, the model allows quantification of how sectoral wage-floor growth and financial frictions (collateral constraints) interact to affect firm outcomes.

The model is of a small open economy within a currency union. The home economy consists of a representative worker, a perfectly competitive retailer and wholesaler, and a continuum of entrepreneurs, $i \in [0, 1]$. The worker consumes hand-to-mouth and supplies labor to entrepreneurs. Monopolistically competitive entrepreneurs produce differentiated intermediate goods and differ in their level of constant idiosyncratic productivity. They hire workers via monopsonistically competitive labor markets, but are potentially constrained in this power by the existence of a wage floor. Entrepreneurs can save and borrow in the international financial market, subject to a collateral constraint. Perfectly competitive wholesalers purchase and aggregate the entrepreneurs' varieties into a tradeable domestic intermediate. The tradeable domestic intermediate can then be sold abroad or to domestic retailers, who then combine it with an imported good to produce the final consumption good.

3.1 Workers

The representative worker consumes hand-to-mouth[8] and chooses the allocation of labor $\{n_{it}\}$ across entrepreneurs. They take rationing constraints set by the entrepreneur $\{\bar{n}_{it}\}$ and wages $\{W_{it}\}$ as given. Workers have iso-elastic preferences over consumption and a convex disutility of labor.

[7]While Italy lacks a nationally set minimum wage, the Italian system is based on contractual wage floors agreed through a centralized collective bargaining process, which applies to all private sector employees. These contractual wages are set at the national level and represent an important source of variation in labor costs to firms, just like changes to a minimum wage. Unlike a national minimum wage, however, collective bargaining agreements set an entire array of contractual wage floors that vary by job title. As a consequence, the direct effects of an increase in contractual wages are not necessarily fully concentrated in the bottom part of the wage distribution. This helps justify the use of a representative worker and the abstraction from distributional effects in my model.

[8]This assumption is made solely for tractability.

Worker preferences are given by

$$\mathcal{U} = \sum_{t=0}^{\infty} \beta^t u(C_t, N_t) = \sum_{t=0}^{\infty} \beta^t \left[\frac{(c_t^w)^{1-\gamma}}{1-\gamma} - \frac{1}{\bar{\varsigma}^{\frac{1}{\varsigma}}} \frac{N_t^{1+\frac{1}{\varsigma}}}{1+\frac{1}{\varsigma}} \right] \tag{3}$$

where $N_t = \left[\int n_{it}^{\frac{\theta+1}{\theta}} di \right]^{\frac{\theta}{\theta+1}}$.

The elasticity of substitution θ governs the ability of workers to move between entrepreneurs. As $\theta \to \infty$, labor market frictions approach zero and entrepreneurs become perfect substitutes. In this case, the household will send workers only to the entrepreneur offering the highest wage.

Workers earn nominal labor income $\int_i W_{it} n_{it}$ and are assumed to be hand-to-mouth. Their consumption satisfies

$$P_t c_t^w = \int W_{it} n_{it} di. \tag{4}$$

The worker chooses labor allocations n_{it} and consumption c_t^w to maximize utility (3) subject to (4) and a rationing constraints set by each entrepreneur, $n_{it} \leq \bar{n}_{it}$. Under certain levels of the wage floor, the labor market for an individual firm might not clear: At high enough values, there may be excess supply. Thus, each entrepreneur is allowed to set a rationing constraint, specifying the maximum amount of labor, $\bar{n}_i$, it is willing to hire under the prevailing wage floor. Let λ_t be the multiplier on the budget constraint. Following Berger et al. (2025), I write the multiplier on the rationing constraint as $\xi_{it} = \lambda_t W_{it}(1 - \nu_{it})$. This allows the first order condition for labor supply to equate the wage multiplied by the normalized multiplier ν_i to the standard product of the worker's marginal rates of substitution,

$$\frac{W_{it}}{P_t} \nu_{it} = \left(\frac{n_{it}}{N_t} \right)^{\frac{1}{\theta}} \left(\frac{N_t}{\bar{\varsigma}} \right)^{\frac{1}{\varsigma}} (c_t^w)^{\gamma}. \tag{5}$$

Complementary slackness requires that $\lambda_t W_{it}(1 - \nu_{it})(\bar{n}_{it} - n_{it}) = 0$. Thus, $\nu_{it} = 1$ when the rationing constraint does not bind, and $\nu_{it} \in (0, 1)$ when the rationing constraint binds. Therefore, when the rationing constraint binds under high enough levels of the wage floor, ν_i provides a wedge between the price paid for labor and the worker's marginal rates of substitution, capturing the inefficiency created by the oversupply of labor.

Equation 5, yields an inverse labor supply schedule

$$w\left(n_{it}, \bar{n}_{it}, \mathbf{S}_t\right) = \begin{cases} \left(\frac{n_{it}}{N_t}\right)^{\frac{1}{\theta}} \left(\frac{N_t}{\bar{\varsigma}}\right)^{\frac{1}{\varsigma}}, & n_{it} \in [0, \bar{n}_{it}) \\ \in \left[\left(\frac{n_{it}}{N_t}\right)^{\frac{1}{\theta}} \left(\frac{N_t}{\bar{\varsigma}}\right)^{\frac{1}{\varsigma}}, \infty\right), & n_{ijt} = \bar{n}_{ijt} \end{cases} \tag{6}$$

Taking as given aggregates $\mathbf{S}_t$, Equation (6) gives the real wage $w_{it} \equiv W_{it}/P_t$ an entrepreneur must pay to the household when it chooses to hire n_{it} workers and set rationing constraint $\bar{n}_{it}$. An entrepreneur will never pay more than the minimum wage necessary to deliver $\bar{n}_{it}$ workers, and thus, $w(n_{it}, \bar{n}_{it}, \mathbf{S}_t)$ is a one-to-one function over $n_i \in [0, \bar{n}_i]$.

3.2 Production

Production of the final consumption good takes place over three stages: First, overlapping generations of entrepreneurs produce differentiated varieties of intermediate goods using capital and labor. They compete monopolistically in the product market and monopsonistically in the labor market. However, monopsonistic power may be limited by the presence of a wage floor. Additionally, entrepreneurs have access to foreign financing that is limited by a collateral constraint. Perfectly competitive wholesalers purchase the entrepreneurs' varieties and combine them into a tradeable intermediate good. Some of these intermediate goods are exported, and the remainder are sold to perfectly competitive retailers. Retailers combine the aggregate intermediate good with imports from abroad to produce a final, non-tradeable good. This good is sold to workers or entrepreneurs. I discuss the production chain in reverse, starting with retailers.

3.2.1 Retailers

Perfectly competitive retailers purchase intermediate goods, D_t, from domestic wholesalers at the nominal price P^Q. Retailers also purchase intermediate goods from abroad, M_t, whose price in foreign currency is fixed and normalized to unity. Converted into domestic currency, the price of imported intermediate goods is $1/E_t$, where E_t is the nominal exchange rate, defined as the price of the home currency in euros. Since I am modeling Italy, I assume the small open economy is part of the euro currency area, so $E_t = 1$. The retailers then assemble the non-tradeable final good, Y_t, which they sell to entrepreneurs or workers at price P_t. Their maximization problem is

$$\max_{D_t, M_t} \left\{ P_t Y_t - P_t^Q D_t - M_t \right\}$$

subject to the production function

$$Y_t = \left\{ \omega^{\frac{1}{\omega}} D_t^{\frac{\psi-1}{\psi}} + (1-\omega)^{\frac{1}{\psi}} M_t^{\frac{\psi-1}{\psi}} \right\}^{\frac{\psi}{\psi-1}}.$$

Here, ψ describes the elasticity of substitution between the domestic and imported intermediate goods, and ω describes the home bias. The optimal demand for the domestic intermediate and imported intermediate goods is given by

$$D_t = \omega Y_t \left(\frac{P_t}{P_t^Q} \right)^{\psi} \tag{7}$$

and

$$M_t = (1-\omega) Y_t P_t^{\psi}.$$

Similarly, the foreign country demands some of the home country's intermediate good. I assume that foreign demand takes the same form as equation (7) so that the demand schedule for the wholesaler's good abroad is

$$X_t = X \left(P_t^Q \right)^{-\psi}, \tag{8}$$

where X is the steady-state level of exports.

3.2.2 Wholesalers

The perfectly competitive wholesaler purchases differentiated varieties $\{q_t(i)\}_{i\in[0,1]}$ from the monopolistically competitive entrepreneurs at prices $\{\varphi_t(i)\}_{i\in[0,1]}$. The varieties are aggregated via a CES production function to produce the tradeable intermediate good, Q_t. The intermediate good is then sold at P_t^Q either domestically to retailers or exported abroad. Wholesalers choose inputs to maximize profits,

$$\max_{q_t(i)} P_t^Q Q_t - \int_0^1 \varphi_t(i) q_t(i) di$$

subject to the CES production function

$$Q_t = \left(\int_0^1 q_t(i)^{\frac{\sigma-1}{\sigma}} di \right)^{\frac{\sigma}{\sigma-1}},$$

where σ is the elasticity of substitution between varieties. This problem yields a standard demand function for each variety,

$$q_t(i) = \left(\frac{\varphi_t(i)}{P_t^Q} \right)^{-\sigma} Q_t. \tag{9}$$

The nominal price of the intermediate good, P_t^Q, is a combination of the prices of the varieties: $P_t^Q = \left(\int (\varphi_t(i))^{1-\sigma} di \right)^{1/(1-\sigma)}$.

3.2.3 Entrepreneurs

Overlapping generations of entrepreneurs consume final goods and produce differentiated varieties of intermediate goods, q_i, using capital and labor. They face monopolistic competition in the product market and monopsonistic competition in the labor market.

At birth, entrepreneurs draw constant idiosyncratic productivity z from a log-normal distribution with p.d.f $f(z)$, mean μ_z, and standard deviation σ_z. They are born with zero debt and receive a transfer of capital from the government $\underline{k}(z)$[9] in order to start production. Entrepreneurs produce differentiated varieties by combining capital and labor via a Cobb-Douglas production function, $q_{it} = z_i k_{it}^{\alpha} n_{ti}^{1-\alpha}$.

An Entrepreneur i has iso-elastic preferences over consumption,

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \frac{(c_{it}^e)^{1-\gamma}}{1-\gamma},$$

where the expectation $\mathbb{E}_0$ is taken over a constant probability of death ρ. At the end of each period, deceased entrepreneurs are replaced by a measure ρ of newborn entrepreneurs. I follow Andreasen et al. (2023) and adopt a standard Blanchard-Yaari formulation, removing the risk of death from the entrepreneur's optimization problem: Entrepreneurs use insurance contracts so that, upon death, all savings and capital are transferred to existing entrepreneurs. Surviving entrepreneurs receive a transfer that increases their net worth by a proportion $\frac{\rho}{1-\rho}$.

Capital depreciates at a rate δ and investment is denoted x_{it}. Taking insurance

[9] $\underline{k}(z)$ varies with z and is set such that all entrepreneurs start with capital at an equal proportion below their optimal scale.

payments into account,[10] the law of motion of capital is given by

$$k_{it+1} = \frac{1}{1-\rho}\left((1-\delta)k_{it} + x_{it}\right).$$

Entrepreneurs also have access to the global market for one period, risk-free nominal discount bonds. d_{it+1} denotes debt issued (bonds sold) at price $\frac{1}{1+i^*}$ where i^* is the world nominal interest rate. They face a standard collateral constraint, constraining the amount borrowed to a fraction $\chi \in [0,1]$ of the value of their capital stock:

$$\frac{d_{it+1}}{P_t(1+i^*)} \leq \chi k_{it+1}. \tag{10}$$

The entrepreneur's budget constraint is

$$P_t c_{it}^e + P_t\left[(1-\rho)k_{it+1} - (1-\delta)k_{it}\right] + d_t + W_{it}n_{it} = \varphi_{it}q_{it} + (1-\rho)\frac{d_{it}}{(1+i^*)} - T_t(z_i),$$

where $T_t(z_i) = P_t\rho\underline{k}(z_i)$ are lump-sum taxes collected by the government to pay for the capital endowments of newborn firms. To avoid the redistribution of income from high- to low-productivity entrepreneurs, the lump-sum tax is specific to type z.

Under monopsonistic labor market competition, the entrepreneur sets the wage, understanding that $\partial W_i/\partial n_i \neq 0$. However, the entrepreneur's choice in wage (and thus the extent of its monopsony power) is constrained by the existence of a nationally set wage floor, $\underline{w}$,

$$W_{it} \geq \bar{w} \geq 0.$$

Due to the wage floor, the labor market for an individual entrepreneur may not clear: For certain levels of $\underline{w}$, workers will want to supply more labor than the entrepreneur demands. To account for this, I follow Berger et al. (2025) and allow each entrepreneur to specify a self-imposed rationing constraint[11] on hiring $n_i \leq \bar{n}_i$.

Recursive formulation of the entrepreneur's problem Following Buera and Moll (2015) and Andreasen et al. (2023), the entrepreneur's recursive optimization problem can be written with cash-on-hand as the only relevant endogenous state variable. Cash on hand is defined as

$$\mathcal{M}_i \equiv \frac{1}{P}\left(\frac{(\varphi_i)^{1-\sigma}}{(P^Q)^{-\sigma}}Q - W_i n_i + P(1-\delta)k_i - d_i - T(z_i)\right).$$

[10] Entrepreneurs accumulate capital by the typical amount, $(1-\delta)k_{it} + x_{it}$ and receives and additional amount via the insurance payouts, $\frac{\rho}{1-\rho}[(1-\delta)k_{it} + x_{it}]$. Together, this yields $k_{it+1} = \frac{1}{1-\rho}[(1-\delta)k_{it} + x_{it})]$.

[11] Berger et al. (2025) explain the rationing constraint as "a sign on the firm's door telling the household the maximum amount of labor the firm is willing to hire."

Then, defining net worth as $a' \equiv k'_i - \frac{d'_i}{EP(1+i^*)}$, the entrepreneur's budget constraint can then be written as

$$c^e_i = \mathcal{M}_i - (1-\rho)a'_i.$$

Using this, the entrepreneur's optimization problem can then be characterized as the solution to a two-stage budgeting problem: The entrepreneur first optimally chooses net worth a'_i to maximize lifetime utility and then makes a static choice to maximize $\mathcal{M}'_i$ by allocating net worth between capital k_i and debt d_i and choosing the variety price φ'_i, employment n_i, and wage W_i. For $t > 0$,[12] an entrepreneur with $(\mathcal{M}, z)$ solves

$$v(\mathcal{M}, z) = \max_{a'} \left[u\left(\mathcal{M} - (1-\rho)a'\right) + \tilde{\beta} v(\tilde{\mathcal{M}}'_i(a', z), z) \right] \tag{11}$$

$$\tilde{\mathcal{M}}'(a', z) = \max_{k', d', \varphi', n', \bar{n}', W'} \frac{1}{P'} \left[\frac{(\varphi')^{1-\sigma}}{(P^{Q\prime})^{-\sigma}} Q' - W'n' - P'(i^* + \delta)k' + P'(1+i^*)a' - T(z) \right] \tag{12}$$

$$\text{subject to} \quad \left(\frac{\varphi'}{P^{Q\prime}}\right)^{-\sigma} Q' = z(k')^{\alpha}(n')^{1-\alpha}, \quad (1-\chi)k' \leq a'$$
$$W' \geq \underline{w}, \quad n' \leq \bar{n}', \quad W'/P' = w\left(n', \bar{n}', \mathbf{S}\right),$$

where $\tilde{\beta} = (1-\rho)\beta$ is the discount factor accounting for the probability of death. Solving the first stage problem yields a decision rule $a'(\mathcal{M}, z)$, describing the evolution of net worth as a function of cash-on-hand. The second stage determines $\tilde{\mathcal{M}}$, optimal allocations capital k, debt d, employment n, and optimal prices φ and W, all as recursive functions of (a', z). Thus, evaluating a' at the optimal value given by $a'(\mathcal{M}, z)$, the second stage decision rules can also be expressed as functions of $\mathcal{M}$:[13] $\tilde{\mathcal{M}}'(\mathcal{M}, z)$, $k'(\mathcal{M}, z)$, $d'(\mathcal{M}, z)$, $\varphi'(\mathcal{M}, z)$, and $n'(\mathcal{M}, z)$.

3.3 Stationary equilibrium

This paper compares stationary equilibria at varying levels of the wage floor. For a given level of the wage floor and assuming $E = 1$, the model's recursive stationary equilibrium consists of aggregate prices $\{P, P^Q\}$, aggregate allocations $\{Q, X, D, M, Y, N\}$, entrepreneurs' decision rules $\{c^{e\prime}(\mathcal{M}, z)$, $a'(\mathcal{M}, z)$, $n'(\mathcal{M}, z)$, $\bar{n}(\mathcal{M}, z)$, $\varphi'(\mathcal{M}, z)$, $q'(\mathcal{M}, z)$, $d'(\mathcal{M}, z)$, $k'(\mathcal{M}, z)$, $W'(\mathcal{M}, z)\}$, worker decision rules $\{c^w, n^s(\mathcal{M}, z)\}$, lump-sum taxes

[12] The problem for $t = 0$ is different because entrepreneurs cannot choose d_0 and k_0; they start with $d_0 = 0$ and are endowed with $k_0 = \underline{k}$ via the government transfer. Thus, in the initial period, the entrepreneur takes these allocations as given and chooses φ_i, n_i, and W_i to maximize cash on hand.

[13] All decision rules also depend upon aggregate variables $\mathbf{S}' = (Y', P', Q', P^{Q\prime}, N',)$. However, since this paper focuses on stationary equilibria where aggregates are time-invariant, I omit $\mathbf{S}$ from the notation for simplicity.

$T(z)$, and value functions $v(\mathcal{M}, z)$ such that

1. Entrepreneurs' value functions and decision rules solve their optimization problem.
2. Decision rules for the demand of intermediate good varieties solve the wholesaler's problem.
3. Decision rules for domestic demand of the aggregate intermediate, the imported intermediate, and output of final goods solve the retailer's problem.
4. The decision rules that characterize labor supply for each entrepreneur solve the worker's problem.
5. The government budget constraint holds: $P\rho\underline{k}(z) = T(z)\ \forall z$.
6. The labor market clear $N = \left[\int n(\mathcal{M}, z)^{\frac{\theta+1}{\theta}} d\phi(\mathcal{M}, z)\right]^{\frac{\theta}{\theta+1}}$
7. The market for final goods clears:

$$\begin{aligned} Y_t =& c^w + \int \rho\underline{k}(z) f(z) \\ &+ \int \left[c^e(\mathcal{M}, z) + (1-\rho)k'(\mathcal{M}, z) - (1-\delta)k(\mathcal{M}, z)\right] \phi(\mathcal{M}, z). \end{aligned}$$

8. The market for the aggregate intermediate good clears: $X_t + D_t = Q_t$.
9. The distribution of firms over $(\mathcal{M}, z)$ is stationary:

$$\phi(\mathcal{M}', z') = \iint \left[(1-\rho)\mathbf{1}^{\mathrm{S}}(\mathcal{M}', \mathcal{M}, z) + \rho\mathbf{1}^{\mathrm{D}}(\mathcal{M}', \mathcal{M}, z)\right] \phi(\mathcal{M}, z) d\mathcal{M} dz,$$

where $\mathbf{1}^{\mathrm{S}}(\mathcal{M}', \mathcal{M}, z)$ and $\mathbf{1}^{\mathrm{S}}(\mathcal{M}', \mathcal{M}, z)$ are dummy variables indicating surviving and deceased firms respectively.

4 Characterization of firm behavior under wage floors

I describe how the wage floor affects entrepreneurs in partial equilibrium and how this effect depends upon the existence of financial frictions. Focusing on the entrepreneur's second-stage problem allows for an analytic exploration of firm behavior before quantifying the full model.

4.1 Preliminaries

Following Berger et al. (2025), I begin by defining three regions of the entrepreneur's problem, for which I will derive optimality conditions. Under successively higher levels of the wage floor $\underline{w}$, an entrepreneur moves across the following regions:

1. Region I: Entrepreneur is unconstrained by w, household is on its labor supply curve.
2. Region II: Entrepreneur is constrained by w, household is on its labor supply curve.
3. Region III: Entrepreneur is constrained by w, household is off its labor supply curve.

An entrepreneur's region of constraint under a given wage floor will ultimately depend on both its net worth a and its productivity z.

4.2 Static effects of wage floors

I now explore the solution of a given entrepreneur's second-stage problem of maximizing $\tilde{\mathcal{M}}_i$ by choosing d_i, k_i, n_i, $\bar{n}_i$, W_i, and φ_i for a given net worth a_i. Since dynamic effects (changes in a) and general equilibrium effects (changes in P, Y, P^Q, Q, N, c^w) are not considered in this section, these can be viewed as the "static" effects of wage floors. The first order conditions of entrepreneur i's second stage problem yield:

$$MRPK_i \equiv \frac{\varphi_i}{\vartheta}\alpha z_i \left(\frac{k_i}{n_i}\right)^{\alpha-1} = P(i^* + \delta) + \eta_i(1-\chi) \tag{13}$$

$$MRPL_i \equiv \frac{\varphi_i}{\vartheta}(1-\alpha) z_i \left(\frac{k_i}{n_i}\right)^{\alpha} = \frac{1}{\mu_i} W_i \tag{14}$$

$$\frac{d_i}{P(1+i^*)} = k_i - a_i \tag{15}$$

$$\varphi_i = \left[\frac{(P^Q)^\sigma Q}{z_i k_i^\alpha n_i^{1-\alpha}}\right]^{\frac{1}{\sigma}} \tag{16}$$

$$\eta\left(a_i - k_i(1-\chi)\right) = 0. \tag{17}$$

where $\vartheta = \sigma/(1-\sigma)$ is the markup of the price over marginal cost, η is the multiplier on the collateral constraint, and $\mu \equiv W_i/MRPL_i$ is the wage markdown charged by the entrepreneur. The size of an entrepreneur's markdown (and if it can charge one at all) will depend upon its region of constraint under the prevailing wage floor.

The marginal revenue product of capital varies across firms and may deviate from the factor price due to the collateral constraint. When $\eta > 0$, the collateral constraints bind and the entrepreneur's choice of capital is pinned down by its net worth, $k(a) = a/(1-\chi)$. Thus, an entrepreneur may be below its optimal scale and have $MRPK_i > P(i^* + \delta)$. Due to the existence of the wage floor, the relationship between the marginal revenue product of labor and the wage also varies across entrepreneurs. This relationship depends on which of the three regions described above the entrepreneur falls into under the prevailing wage floor. With no wage floor, all firms would exercise their monopsony power to pay a wage equivalent to a constant markdown over the marginal revenue product of labor. Thus, the entrepreneur's labor optimality condition must be described region by region:

Region I: In this region, the entrepreneur is unconstrained by w and the worker is on their labor supply curve. Under a scenario with no wage floor, all firms would fall into region I. In this region, the entrepreneur behaves as a standard monopsonist: Rationing constraints are irrelevant, and wages are a fixed markdown over the marginal revenue product of labor

$$MRPL_i = W_i + \frac{\partial W_i}{\partial n_i} n_i = W_i \left(1 + \frac{\partial \log W_i}{\partial \log n_i}\right) = W_i(1 + \frac{1}{\theta}).$$

In region I, the entrepreneur retains its full monopsonistic power, allowing it to pay workers a markdown μ_i over their marginal product of labor given by $\frac{\theta}{1+\theta}$. Since the household remains on its labor supply curve in this region, the rationing constraint does not bind and $\nu_i = 1$. Thus, the entrepreneur's labor allocation can be read off the worker's labor supply curve, and its wage, markdown, and employment are given by

$$W_i = \frac{\theta}{1+\theta} MRPL_i, \quad \mu_i = \frac{\theta}{1+\theta}, \quad \text{and} \quad n_i' = \left(\frac{W_i}{P}\left(\frac{N}{\bar{\varsigma}}\right)^{\frac{-1}{\varsigma}} (c^w)^{-\gamma}\right)^{\theta} N.$$

Region II: In region II, the entrepreneur is constrained by the wage floor, but the rationing constraint does not bind, so the worker is still on the labor supply curve. Thus,

$$W_i = \underline{w}, \quad \mu_i = \frac{\underline{w}}{MRPL_i} \quad \text{and} \quad n_i' = \left(\frac{w_i}{P}\left(\frac{N}{\bar{\varsigma}}\right)^{\frac{-1}{\varsigma}} (c^w)^{-\gamma}\right)^{\theta} N.$$

For an entrepreneur moving within region II, as the wage floor increases, μ_i increases, narrowing the markdown the firm can charge. At the border of regions I and II, the entrepreneur acts as a full monopsonist and charges $\mu_i = \theta/(1+\theta)$. At the border of regions II and III, the wage and the marginal product of labor are equalized, and thus $\mu_i = 1$, so that the level of the individual entrepreneur's employment allocation is efficient.

Essentially, as an entrepreneur moves through region II, its monopsony power is slowly depleted until it behaves as it would in a competitive labor market.

Region III: In region III, the firm is constrained by the wage floor, and the rationing constraint binds such that $\nu_i < 1$ and the worker is off their labor supply curve. In this region, the entrepreneur no longer has any power to charge a markdown, the wage floor pins down the wage, and employment is at the maximum level determined by the rationing constraint,

$$W_i = \underline{w} = MRPL_i, \quad \mu_i = 1, \quad \text{and} \quad n_i = \bar{n}_i.$$

While the entrepreneur has no power to charge a markdown in this region, this does not mean there is no inefficiency at high wage floors: the entrepreneur's labor market no longer clears, resulting in inefficiency from the binding rationing constraint. When $\nu < 1$, there is a wedge between the wage paid and the worker's marginal rates of substitution. As the wage floor rises and the entrepreneur moves further into region III, the rationing constraint binds more strongly, ν approaches zero, and the associated inefficiency grows.

Rationing constraints: To solve for the optimal rationing constraint, note that entrepreneurs will never hire at a point where the marginal revenue product of labor is below the wage floor. Thus, it is weakly optimal to set the rationing constraint by finding $\bar{n}_i$ such that $MRPL_i/P = \underline{w}$. Using the entrepreneur's optimality conditions, this yields

$$\bar{n}_i = \left(\frac{1-\alpha}{\vartheta} \frac{P^Q}{\underline{w}} (Q)^{\frac{1}{\sigma}} (z_i k_i^{\alpha})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\alpha\sigma+1-\alpha}}. \tag{18}$$

Determining a firm's region: To determine a firm's region under $\underline{w}$, start with an initial guess that the firm is in region II. Under this assumption, compute the entrepreneur's markdown:

$$\hat{\mu}_i^{II} = \frac{\bar{w}}{MRPL_i^{II}}. \tag{19}$$

If $\hat{\mu}_i^{II} > 1$, then the firm is in region III and if $\tilde{\mu}^{II} < \frac{\theta}{1+\theta}$, then the firm is in region I. Otherwise, the guess was correct, and the firm lies in region II. In the absence of binding collateral constraints and taking aggregates as given, $\hat{\mu}_i^{II}$ (and therefore the corresponding region) will depend only on an entrepreneur's productivity, $\hat{\mu}_i^{II}(z_i)$, for a given $\underline{w}$. In this case, there will exist $\bar{z}$ and $\underline{z}$ such that $\bar{z} > \underline{z}$ and any entrepreneur with $z_i < \underline{z}$ will be in region III and any entrepreneur with $z_i > \bar{z}$ will be in region I. However, for an entrepreneur facing binding collateral constraints, the distinction between regions will

depend jointly on productivity and net worth, $\hat{\mu}^{(2)}(z_i, a_i)$. Thus, two entrepreneurs of the same z_i may no longer share the same region if they differ in net worth.

4.2.1 Static effects with no financial frictions

To create a baseline, I first explore the static effects of a wage floor in an environment with no financial frictions. To remove financial frictions from the model, assume $\chi \to \infty$ so that the collateral constraint never binds. In this case, all firms equate their marginal revenue products of capital:

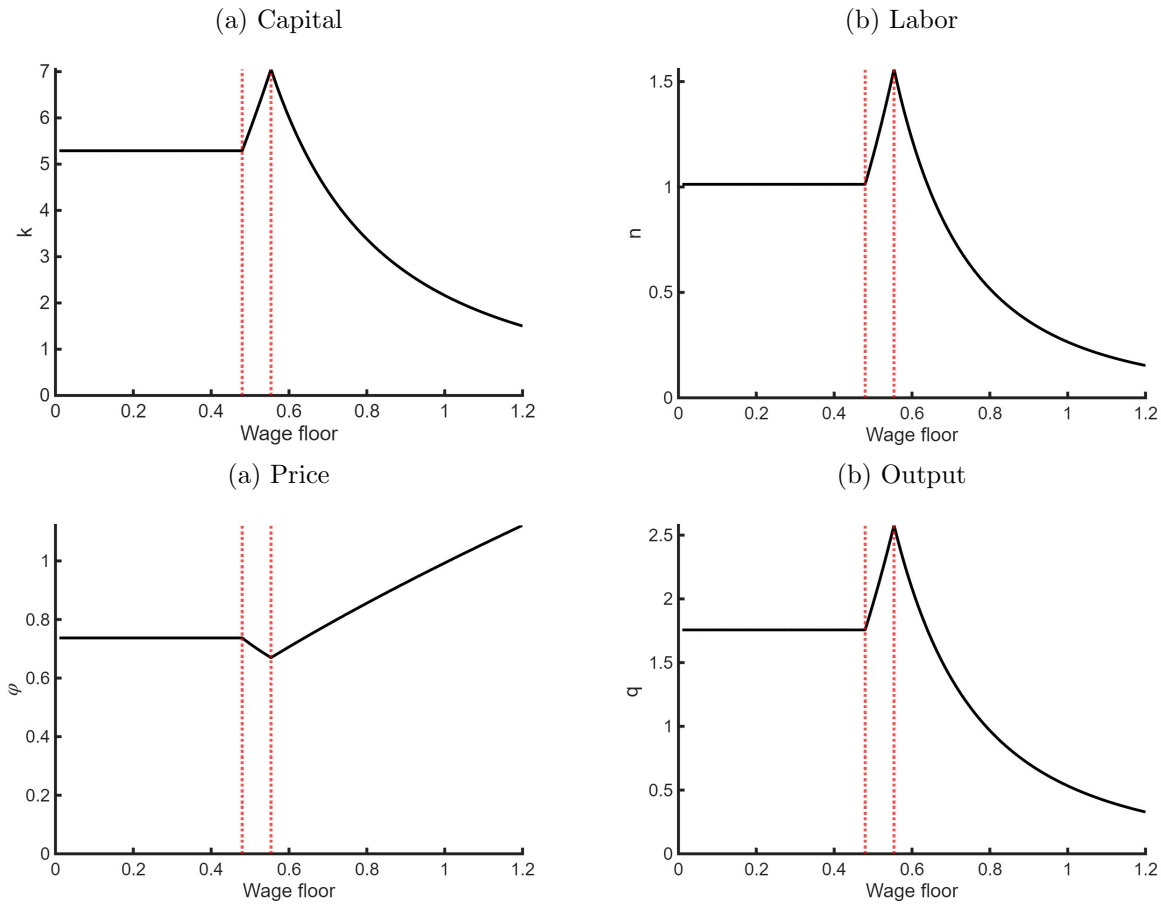
$$MRPK_i = P'(i^* + \delta) \quad \forall i. \tag{20}$$

In the absence of financial frictions, there is no dispersion in $MRPK$ across firms: If an entrepreneur's initial capital allocation, $\underline{k}(z_i)$, is below its steady state value $k^*(z_i)$, the entrepreneur will borrow however much is needed to jump immediately to its optimal scale. Thus, in solving the second stage problem, net worth becomes irrelevant. The constant marginal revenue product of capital in this environment pins down the capital-labor ratio. Using this ratio, the entrepreneur's marginal revenue product of labor depends only on z_i. Applying this insight to Equation 19, the entrepreneur's region under a given wage floor then depends only on z_i. Thus, an entrepreneur's region of constraint under the given wage floor and allocations of capital k_i and labor n_i, and thus its production q_i and price φ_i, depend only on productivity z_i. Full mathematical details are provided in Appendix B.3.

Figure 4 plots the static trajectories of capital, labor, price, and output for an entrepreneur of a given z_i as a function of the wage floor. The dashed lines divide the entrepreneur's regions of constraint. At low levels of the wage floor, from $\underline{w} = 0$ up until the first dashed line when the entrepreneur enters region II, allocations of capital and labor are constant with respect to $\underline{w}$. In region I, the wage floor is below the wage the monopsonist optimally charges and thus has no static effect on the entrepreneur's decisions. The worker's labor supply curve determines the allocation of labor, and the capital allocation satisfies Equation 20.

In region II, for levels of the wage floor between the two dashed lines, capital and labor allocations increase with $\underline{w}$. In this region, the wage floor is above the wage optimally set by the monopsonist but low enough such that the rationing constraint does not bind. The entrepreneur must set wages equal to $\underline{w}$. The employment allocation can be read off the worker's labor supply curve, so as w_i increases to match $\underline{w}$, employment increases as well. As labor allocations increase, capital adjusts up via factor complementarity to

Figure 4: Static effects of wage floors with no collateral constraint



Note: The figure plots the static effects of increasing the wage floor on capital, labor, price, and output for an entrepreneur of a given z_i in an environment with no collateral constraints. The red dashed lines delineate between regions of constraint with respect to the wage floor: Region I is the far left portion of the graph, Region II falls between the two dashed lines, and Region III is the far right portion. Dynamic and general equilibrium effects are not considered.

maintain the capital-labor ratio determined by 20. Capital and labor allocations peak at the border of Regions II and III: At this level of $\underline{w}$, the wage and the entrepreneur's marginal revenue product of labor are equalized, such that the employment allocation is efficient at the level of the individual firm. Here, the wage floor is high enough to deplete the entrepreneur's full monopsony power, but low enough to avoid over-supply of labor and associated market inefficiencies.

In region III, for all levels of the wage floor above the second dashed line, capital and labor allocations decrease with $\underline{w}$. In this region, the wage floor is high enough such that the entrepreneur's rationing constraint binds. The entrepreneur's monopsony power is fully depleted at the border between regions II and III, so any further wage floor increases have no scope to improve efficiency from the perspective of the individual entrepreneur. The entrepreneur will never hire labor such that $MRPL_i < W_i = \underline{w}$. Thus, in region III, as the wage floor increases, employment decreases to keep the marginal revenue product

of labor from falling below the wage floor. As employment decreases, capital adjusts down via factor complementarity. Moving through region III, capital and labor allocations eventually fall below the allocations of the full monopsonist (region I) and, in the limit, both approach zero.

4.2.2 Static effects with financial frictions

I now consider the static effects of wage floors in the full environment with collateral constraints. Under a binding collateral constraint, an entrepreneur will be unable to borrow and thus may be constrained at a level of capital below its optimal scale:

Proposition 1: For χ sufficiently low such that the collateral constraint binds for entrepreneur i, $MRPK_i > \overline{MRPK}$ and $k_i < \bar{k}_i$, where $\bar{k}_i$ is the firms steady state (optimal scale) of capital[14] and $\overline{MRPK} = P(i^* + \delta)$.

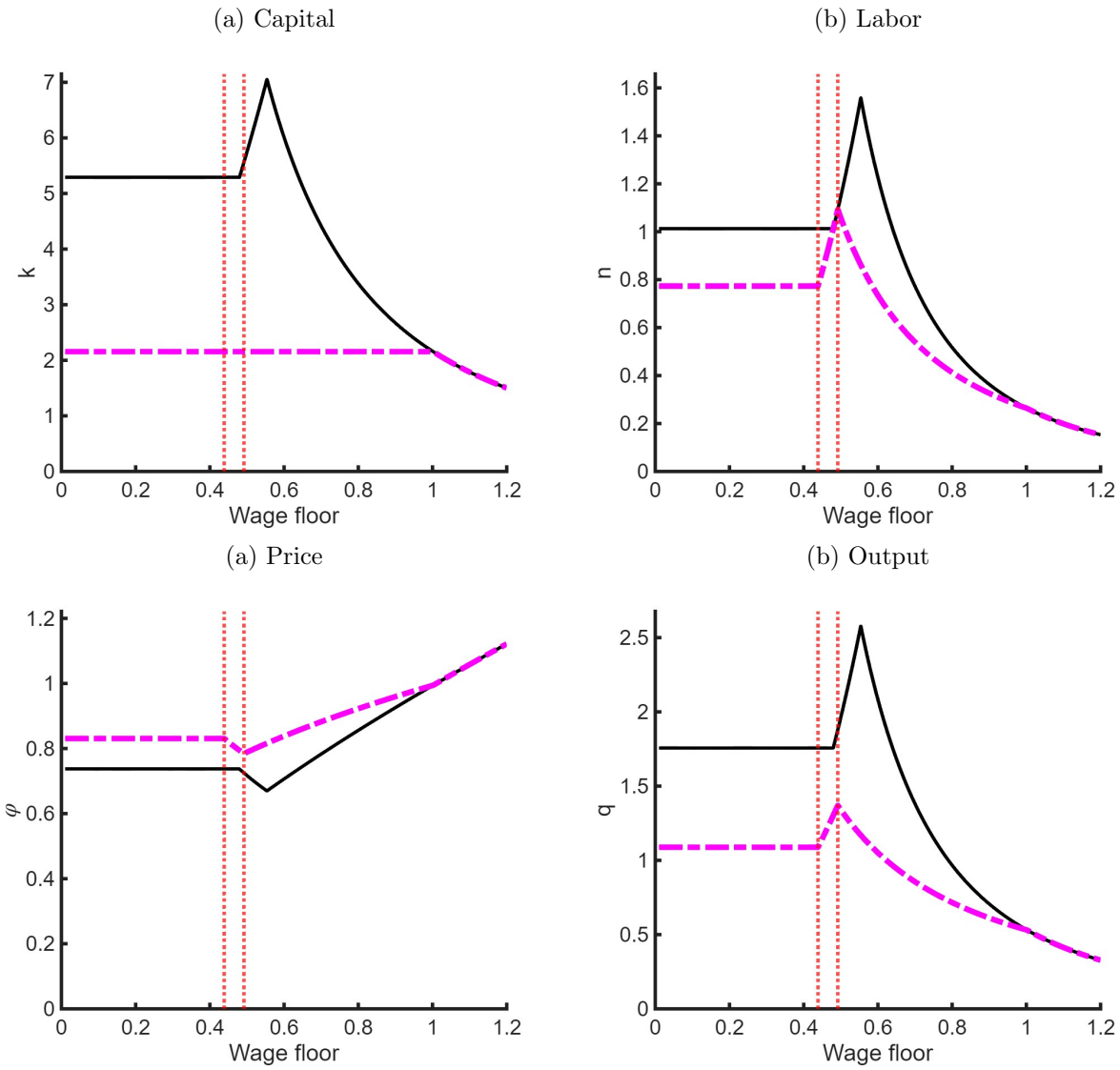
Assuming the collateral constraint binds, an entrepreneur's investment choice depends only on its net worth, regardless of productivity or wage floor, $k(a_i) = a_i/(1-\chi)$. As a result, the optimal choice of labor and the region of constraint under a given wage floor $\underline{w}$ will now depend on both net worth and productivity. Full mathematical details and proofs are available in Appendix B.4.

Figure 5 plots the static trajectories of capital, labor, price, and output for entrepreneurs of a given z_i as a function of the wage floor. The solid black line corresponds to the trajectories of the entrepreneur when the net worth a_i is high enough such that the collateral constraint never binds. These allocations align with those of the entrepreneur in an environment with no financial frictions: In both cases, the entrepreneur operates at its optimal scale. The pink dashed line corresponds to the allocations for the same entrepreneur (same z_i) at a lower level of net worth such that the collateral constraint binds under certain values of $\underline{w}$. The red dashed lines designate the different regions for the constrained (low net worth) entrepreneur. Note that the dashed lines no longer align with the kinks in the allocations for the high net worth firm, reflecting that two entrepreneurs of the same productivity but different net worth may be in different regions under a given wage floor.

When constrained by the collateral constraint, the entrepreneur's capital allocation remains invariant across regions. As long as its optimal scale remains out of reach, the entrepreneur chooses the closest possible allocation and sets $k(a_i) = a_i/(1-\chi)$ regardless of $\underline{w}$. The capital allocation of the low net worth entrepreneur adjusts and aligns with

[14]The steady state level of k_i will depend upon the prevailing wage floor $\underline{w}$ and corresponds to the capital allocations described in the environment with no collateral constraints.

Figure 5: Static effects of wage floors with collateral constraint



Note: The figure plots the static effects of increasing the wage floor on capital, labor, price, and output for an entrepreneur of a given z_i in an environment with collateral constraints. The solid black line represents allocation when the entrepreneur's net worth is high enough such that the collateral constraints never bind. The pink dashed line represents the allocations for that entrepreneur at a low net worth, such that the collateral constraint binds at certain levels of the wage floor. The red dashed lines delineate between regions of constraint with respect to the constrained firm.

the high net worth allocation only at the far right end of region III: Here, wage floors are so high that the optimal scale falls below $a_i/(1-\chi)$, and the entrepreneur escapes the collateral constraint.

The low net worth entrepreneur's labor allocations over $\underline{w}$ mirror those of the unconstrained entrepreneur, but are shifted downward and to the left. The lower level of capital, fixed by the collateral constraint, reduces the marginal revenue product of labor via factor complementarity. Thus, from Equation 19, the constrained entrepreneur

transitions from region to region at lower levels of the wage floor than the unconstrained entrepreneur. Looking at Figure 5, region II is bound by the two red dashed lines for the low net worth entrepreneur and by the two kinds in the black line for the high net worth entrepreneur, with the constrained entrepreneur transitioning into region III just after the unconstrained entrepreneur enters region II. Thus, the wage floor maximizes entrepreneur-level efficiency at lower levels of $\underline{w}$ in an environment with financial frictions and becomes costly sooner, with labor allocations falling below the monopsony (region I) levels at lower $\underline{w}$.

The discussion in this section focuses only on the static effects of wage floors. It considers only one entrepreneur in isolation and abstracts from general equilibrium changes in aggregate prices and quantities, as well as from dynamic effects on net worth. Understanding the full effects of the wage floors requires quantifying the model.

5 Quantitative analysis

In this section, I quantify the model to analyze the full general equilibrium implications of wage floors in the Italian manufacturing sector. I outline the solution method in Appendix C.3. I calibrate the model to Italy in 1999–2000, representing a baseline for the economy before the surge in labor costs.[15] To calibrate the model, I consider the steady state of the economy, so that aggregate variables do not vary with t and the measure of entrepreneurs is stationary. From this baseline calibration, I then increase the model's wage floor[16] by 20 percent to match the rise in labor costs observed in the data between 2000 and 2019. By comparing the stationary equilibria in the two calibrations, I can assess in detail the interaction between firm performance, aggregate outcomes, financial frictions, and the increase in wage floors. In addition, I conduct counterfactual exercises and consider the welfare implications associated with the increase in wage floors.

5.1 Calibration

The model is calibrated to match Italy during the period 1999-2000, prior to the rise in labor costs. I proceed in two main steps: First, I fix a subset of parameters to

[15] Ideally, the initial calibration would be set earlier to avoid potential disruptions from euro area integration. However, Orbis data coverage for Italy before 1999 is limited.

[16] Here, for simplicity, I assume that all manufacturing firms are subject to the same wage floor. In reality, there are many negotiated contracts under the manufacturing umbrella. While these contracts vary in the size and time of their adjustment, they correlated strongly in their overall adjustment from 1999 to 2019 (see the time series in Figure 3). The average wage floor increase for the manufacturing sector between 1999 and 2019 was approximately 20%.

match standard values. Second, I choose the remaining parameters so that the model's baseline stationary equilibrium moments match key features of the data.

Table 4: Fixed parameters

Parameter	Description	Value
β	Discount factor	0.96
γ	Risk aversion	2.00
σ	EoS b/w varieties	3.25
ψ	Armington elasticity	3.75
ς	Frisch elasticity	0.62
ω	home bias	0.70
δ	Depreciation rate	0.06
ρ	Death probability	0.07
θ	Between firm labor elasticity	3.00

Fixed Parameters Standard values from the literature are used to set several of the parameters of the model. Table 4 lists these parameters and their values. I consider a CRRA utility of consumption with relative risk aversion $\gamma = 2$ for both the worker and the entrepreneur. The worker's utility function has a disutility of aggregate labor governed by the Frisch elasticity $\varsigma = 0.62$. The subjective discount factor is set to $\beta = 0.96$. The elasticity of substitution across varieties is $\sigma = 3.25$. Governing trade in the economy, the home bias is set to $\omega = 0.7$, and the Armington elasticity is set to $\psi = 3.75$. The rate of depreciation is set to $\delta = 0.06$ in line with Midrigan and Xu (2014). The death probability is set to the exit rate of firms seen in the Italian data from 1999 to 2019, $\rho = 0.07$. I set the worker's elasticity of substitution between entrepreneurs to $\theta = 3$, generating a markdown of 0.75 for the unconstrained monopsonist.[17] This is in line with the value used in Berger et al. (2025) when considering neoclassical monopsony and in line with empirical estimates for the euro area surveyed in Sokolova and Sorensen (2021).

Fitted Parameters The remaining parameters are calibrated using an SMM algorithm with equal weight on each parameter, minimizing the squared differences of model moments from data targets. The targets are: (1) The ratio of sales of large firms to small firms; (2) Average leverage; (3) Aggregate capital stock as a fraction of the wage bill; (4) The ratio of the labor share (wage bill over value added) of large firms to small firms; and (5) The ratio of investment shares of large firms to small firms. Large firms are those in the top tercile of size, determined by the log of real total assets, and small firms are the

[17]The model is quite sensitive to the joint choice of θ and $\underline{w}$. Since there are empirical estimates for θ, but no clear empirical counterpart for the **level** of $\underline{w}$, I choose to fix θ and estimate $\underline{w}$. The initial level of the wage floor is critical in determining the effects of the 20% increase. The initial level of $\underline{w}$ is contextualized with respect to an "optimal" wage floor in Appendix C.2.

Table 5: Fitted parameters and targets

Parameter	Description	Value	Target (all joint)	Data	Model
σ_z	Prod. dispersion	0.52	Ave. Sales (S_3/S_1)	25.43	25.34
χ	Collateral coefficient	0.09	Ave. leverage	0.26	0.26
α	Capital intensity	0.39	Agg. capital stock / wage bill	5.75	5.78
$\underline{w}_0$	Initial wage floor	0.42	Ave. wage bill/VA (S_3/S_1)	0.84	0.82
κ	Frac. of k^* endowed	0.26	Ave. Inv/VA (S_3/S_1)	1.97	1.98

Notes: Left panel contains the parameters chosen to match the moments in the right table. (S_3/S_1) indicates the ratio of firms by tercile size (log of real total assets). Calibration uses SMM with equal weight on each parameter to jointly target moments.

bottom tercile. The resulting parameter values are reported in Table 5. Data moments are computed using firm-level data from Orbis for 1999-2001. The calibration yields model moments that are quite close to the data moments.

Table 6: Distribution of capital and labor by quintiles

Quintile	**Data 2000**	**Model pre-$\underline{w}\uparrow$**	**Data 2019**	**Model Post-$\underline{w}\uparrow$**
Capital Distribution				
0.2	0.011	0.014	0.004	0.011
0.4	0.033	0.051	0.016	0.023
0.6	0.068	0.083	0.044	0.061
0.8	0.142	0.149	0.117	0.121
1	0.746	0.703	0.819	0.784
Labor Distribution				
0.2	0.028	0.031	0.011	0.021
0.4	0.059	0.086	0.036	0.051
0.6	0.098	0.122	0.072	0.108
0.8	0.167	0.190	0.148	0.160
1	0.649	0.570	0.733	0.660

Validation: Table 6 reports the model's firm-size distribution based on capital and employment in the model for the baseline calibration ("pre-$\underline{w}\uparrow$") and for the outcome of increasing the wage floor by 20 percent ("post-$\underline{w}\uparrow$") alongside equivalent empirical moments for 1999–2000 and 2019. The model's firm-size distributions in the baseline and 20 percent wage floor increase equilibria approximate those observed in the data quite well. Both distributions display significant heterogeneity and a concentration of capital ownership and employment in the top quintile. The fractions of capital held by each quintile of firms in the data and in the model are very similar, with the lowest quintile holding less than 2% of aggregate capital and the top quintile holding about 70-80%. Labor is somewhat less unequally distributed, with the top quintile holding about

60-70%. The combined influence of monopsony power and wage floors plays a central role in the model's ability to match the firm size distribution as well as it does: Small, low productivity entrepreneurs are most likely to be heavily constrained by the wage floor, pushed far into region III where optimal allocations are forced down towards zero. The distribution of entrepreneurs by region of constraint across both calibrations is reported in Appendix C.1.

Changes in the distributions of both capital and labor across time periods and calibrations are relatively muted, showing only a slight shift towards greater inequality following the wage floor increase. In the discussion of static effects, I showed changes in the wage floor can affect firm size in terms of both capital and employment. However, this does not necessarily change how the aggregate capital stock and supply of labor are distributed across entrepreneurs.

5.2 Effects of contractual wage floor increase

Table 7: Aggregate effects of wage floor increase

Variable	20% $\underline{w} \uparrow$	10% $\underline{w} \uparrow$	$\underline{w} = 0$	20% $\underline{w} \uparrow$ & $\hat{\chi} = 2\chi$	20% $\underline{w} \uparrow$ & $\hat{\omega} = 0.6$
	(1)	(2)	(3)	(4)	(5)
Investment	−4.81%	−2.22%	+1.05%	−2.54%	−5.93%
Employment	−6.05%	−3.11%	+1.15%	−3.83%	−7.82%
Entrepreneur sales	−3.54%	−1.04%	+0.62%	−1.15%	−2.85%
Domestic sales	−3.35%	−0.87%	+0.70%	−1.05%	−2.45%
Exports	−3.51%	−1.60%	+1.13%	−2.06%	−4.82%
Labor income (nom)	+0.46%	+0.61%	+0.98%	+0.85%	−0.37%
Price Level	+1.93%	+1.14%	−0.65%	+1.52%	+1.88%
Worker consumption	−1.91%	−0.56%	+1.60%	−0.64%	−2.95%
Entrepreneur consumption	−2.05%	−0.97%	+0.91%	−1.05%	−2.30%
Real GDP	−5.28%	−2.01%	+1.35%	−2.57%	−4.33%

Notes: Percent changes relative to baseline calibration (1999–2000, pre-$\underline{w} \uparrow$). Column (1) presents the baseline results, comparing the equilibria before and after the 20% wage floor increase. Column (2) moderates the wage floor increase to 10%. Column (3) removes the wage floor but keeps monopsony power. Column (4) raises the wage floor by 20% and loosens collateral constraints ($\hat{\chi} = 2\chi$). Column (5) raises the wage floor by 20% and lowers the degree of home bias ($\hat{\omega} = 0.6 < 0.7$).

Column (1) of Table 7 quantifies the effects of a 20% wage floor increase on aggregate variables in the economy. The results are reported as percentage differences from the baseline calibration and reveal significant damage to the Italian economy. The wage floor increase is associated with aggregate contractions in investment (-4.81%), employment (-6.05%), and sales (-3.54%) for entrepreneurs. Higher wages partially pass through,

increasing the price level (+1.93%) and depressing both domestic demand and exports (-3.51%). Real GDP declines by 5.28 percent, with consumption falling for both workers (–1.91%) and entrepreneurs (–2.05%). The negative effect on worker consumption indicates that lower employment and higher domestic prices more than offset wage gains for the worker[18].

Table 8: Heterogeneous effects of 20% $\underline{w} \uparrow$ by size and leverage

Dependent variable	Investment	Employment	Sales
Effects by terciles of leverage			
Bottom tercile	−3.22%	−4.84%	−1.65%
Middle tercile	−5.03%	−6.51%	−2.35%
Top Tercile	−6.85%	−8.21%	−3.10%
Effects by terciles of size			
Bottom tercile	−7.41%	−5.42%	−3.85%
Middle tercile	−4.80%	−5.53%	−2.21%
Top Tercile	+0.95%	−0.82%	+1.30%
Effects by size and leverage			
Small, low leverage	−5.81%	−7.43%	−3.15%
Large, low leverage	+1.15%	−0.22%	+1.05%
Small, high leverage	−9.21%	−10.47%	−4.63%
Large, high leverage	−2.85%	−3.50%	−1.44%

Notes: Table reports the percent change from baseline for different groups of firms under a 20% increase in the wage floor. Size is defined based on the log of real total assets. Leverage is defined as the ratio of debt to assets. Categories for size–leverage interactions follow median splits.

Consistent with the empirical evidence, aggregate contractions mask significant heterogeneity among entrepreneurs by size and leverage, suggesting a role for financial frictions. Table 8 reports the effects of a 20% wage floor increase on aggregate investment, employment, and sales for entrepreneurs across different classes of size and leverage. Looking first at entrepreneurial outcomes by tercile of leverage, highly levered (top tercile) entrepreneurs experience the largest contractions, with investment falling by roughly three times that of low-leverage (bottom tercile) entrepreneurs. Looking by tercile of size, small

[18]Note that the 20% wage floor increase does not map into a 20% increase in nominal wages for the worker. Only entrepreneurs constrained both before and after the wage floor increase experience the full 20% increase in wages. For example, entrepreneurs who remain in region one under both equilibria are unconstrained and unaffected by the wage floor increase, aside from general equilibrium effects. While only a small fraction of large and highly productive entrepreneurs are fully unconstrained, they account for a substantial portion of total employment, as seen in Table 6.

(bottom tercile) entrepreneurs experience large contractions across all three variables. In contrast, large firms see increases in investment and sales–a result of gaining market share as smaller competitors shrink.

Interacting size and leverage, small and highly leveraged entrepreneurs–the most likely to be constrained in borrowing–experience the largest declines in both employment and investment, with investment falling by 9.21 percent and labor by 10.47 percent. As wage bills increase and sales contract, these entrepreneurs are unable to accumulate net worth and remain constrained below their optimal scale. Large and low-leverage entrepreneurs are comparatively insulated and appear to absorb some market share as constrained rivals shrink: Employment contracts slightly (-0.22%), but investment (+1.15%) and sales (+1.05%) expand following the wage floor increase.[19] This reallocation toward large, low-leverage firms helps to temper the negative effects of the wage floor increase but falls far short of overturning it.

Taken together, the baseline quantitative results provide a unified narrative with the empirical evidence: A binding wage floor raises labor costs, lowers net worth, and—through collateral constraints—pushes investment down. The burden falls disproportionately on small and highly leveraged firms, resulting in a reallocation that is quantitatively meaningful but insufficient to offset the aggregate decline in capital, output, and consumption. To clarify the mechanisms, I proceed by decomposing the aggregate contractions and attributing effects to financial or labor market rigidities.

Decomposing the contributions of wage floor increases and financial frictions: Table 9 decomposes the effects of a 20 percent increase in the contractual wage floor into the contributions of wage floor increase, financial frictions, and their interaction. The decomposition requires comparing entrepreneurs' outcomes under three different policy regimes: The first scenario, "20% $\underline{w}\uparrow$, keep CC," represents the baseline reality where firms face both the 20% contractual wage floor increase and a constant level of financial frictions in the form of a potentially binding collateral constraint. The second scenario, "20% $\underline{w}\uparrow$, remove CC," considers a counterfactual outcome where the wage floor increase still occurred, but all borrowing constraints are removed ($\chi \to \infty$). The third scenario, "0% $\underline{w}\uparrow$, remove CC," presents a counterfactual exercise where wage floors remain constant at the 1999 level, and financial frictions are eliminated. By comparing these three scenarios, I can isolate how much damage comes from the increase in wage floors, how much from the existence of collateral constraints, and how much additional damage results from their interaction.

[19] Large firms gaining market share when wage floors are increased is consistent with the results in Lo Bello and Pesaresi (2025).

Table 9: Decomposing financial and labor market distortions

	Sales	Capital	Labor	Welfare
Panel A: Percent change from baseline				
20% $\underline{w}\uparrow$, keep CC	−3.54%	−4.81%	−6.05%	−1.53%
20% $\underline{w}\uparrow$, remove CC	−1.03%	+1.30%	−3.73%	−0.40%
0% $\underline{w}\uparrow$, remove CC	+1.43%	+3.61%	+1.11%	+0.81%
Panel B: Decomposition of effects				
Wage floor effect	−2.45%	−2.30%	−4.84%	−1.21%
Collateral constraint effect	−1.43%	−3.61%	−1.11%	−0.81%
Interaction effect	−1.09%	−2.51%	−1.21%	−0.32%
Total effect	−4.97%	−8.42%	−7.16%	−2.34%

Notes: Panel A shows percent changes relative to baseline case with starting level of wage floors and collateral constraints (CC). The 3 cases are the main experiment of increasing the wage floor by 20% (20% $\underline{w}\uparrow$, keep CC), an experiment where the wage floor is raised by 20% but collateral constraints are removed (20% $\underline{w}\uparrow$, remove CC), and a case where no wage increase occurs but collateral constraints are removed (0% $\underline{w}\uparrow$, remove CC). Panel B decomposes total effects using wage floor effect = (20% $\underline{w}\uparrow$, remove CC) - (0% $\underline{w}\uparrow$, remove CC); Collateral constraint effect = -(0% $\underline{w}\uparrow$, remove CC); Total effect of wage floors and financial frictions = (20% $\underline{w}\uparrow$, keep CC) - (0% $\underline{w}\uparrow$, remove CC). Interaction = Total effect - wage floor effect - collateral constraint effect.

Panel B of Table 9 isolates and quantifies these effects. The wage floor effect measures the damage from the 20 percent wage floor increase when entrepreneurs have perfect capital market access–the difference between outcomes under the 20% wage floor increase and no wage floor increase when financial constraints are absent (scenario 2 - scenario 3). This effect represents the cost of the contractual wage floor increase in a frictionless financial system. Even without financial constraints, the wage floor increase still has significant negative effects on capital (-2.30%), labor (-4.84%), and sales (-2.45%). In an environment without financial frictions, entrepreneurs can always borrow to invest and jump to their optimal scale. However, by raising labor costs and lowering employment, the contractual wage floor increases lower the optimal scale of capital via factor complementarity and thus still lead to a contraction in investment.

The collateral constraint effect measures the cost of financial frictions in the absence of a wage floor increase, capturing the benefit from removing financial frictions relative to the baseline calibration (-scenario 3). As with the wage floor effect, the collateral constraint effect negatively impacts capital (-3.61%), labor (-1.11%), and sales (-1.43%). Investment is suppressed because the collateral constraint prevents entrepreneurs from reaching their optimal scale, limiting them to accumulating capital in proportion to their net worth. Despite eliminating the wage floor increase, employment still contracts due to

factor complementarity.

The interaction effect measures the extent to which the coexistence of financial frictions amplified the effects of the contractual wage floor increase. The interaction effect primarily impacts capital (-2.51%), accounting for about 30% of the total combined impact of wage floor increases and financial frictions (-8.42%). The wage floor increase increases wage bills, cutting profits and slowing net worth accumulation. Thus, with the 20% wage floor increases, firms remain small and constrained: Rather than gradually growing out of borrowing constraints, firms become trapped in low-investment equilibria where inadequate capital makes their (expensive) workforce unproductive. Employment shows a smaller interaction effect (1.21%, or 17% of the total). The 20% wage floor increase has a large negative impact on employment allocations. While collateral constraints can also reduce employment through factor complementarity, the effect is not large enough to significantly compound this impact. The decomposition demonstrates the importance of reforms addressing both financial and labor market rigidities in restoring Italy's competitiveness.

5.3 Welfare effects of wage floors

To assess the effect of the wage floor increase on social welfare, I compute a compensating variation in consumption using a utilitarian social welfare function. Applied to the equilibrium with the 20% wage floor increase, the compensating variation measures the increase in consumption required to make entrepreneurs and workers indifferent to the wage floor increase.

A complete measure of welfare requires that entrepreneurs be weighted equally before and after the wage floor increase. To accomplish this, I follow Andreasen et al. (2023) and make use of the stationary firm-age distribution implied by the model's Blanchard-Yaari structure. Under Blanchard-Yaari, the firm-age distribution is exogenous: $\phi(\tau, z) = \rho(1-\rho)^{\tau} f(z)$. Thus, while entrepreneurs will have different $(\mathcal{M}, z)$ under the two regimes, the fraction of entrepreneurs of a given age and productivity is invariant.[20] The welfare weights then correspond to the fraction of entrepreneurs in a given age and productivity cohort in the stationary equilibrium, $\phi(\tau, z)$. The compensating differential is defined as

$$G = \left[\frac{\sum_{\tau} \sum_{z} V^{\underline{w}\uparrow}(\tau, z) \phi(\tau, z)}{\sum_{\tau} \sum_{z} V^{\text{base}}(\tau, z) \phi(\tau, z)} \right]^{\frac{1}{1-\gamma}} - 1. \tag{21}$$

When considering welfare measures for entrepreneurs of a certain class (e.g., leverage

[20] Using the decision rules $a'(\mathcal{M}, z)$ and $\mathcal{M}'(a, z)$, the solutions obtained over $(\mathcal{M}, z)$ can be mapped into the (τ, z) space via recursive substitution starting from the newborn entrepreneur's choice $a'(\tau = 0, z) = a'(\underline{\mathcal{M}}(z), z)$ and $\mathcal{M}'(\tau = 0, z) = \mathcal{M}'(a'(\tau = 0, z), z)$.

Table 10: Welfare analysis

Variable	20% $\underline{w}\uparrow$	10% $\underline{w}\uparrow$	$\underline{w}=0$	20% $\underline{w}\uparrow$ & $\hat{\chi}=2\chi$	20% $\underline{w}\uparrow$ & $\hat{\omega}=0.6$
	(1)	(2)	(3)	(4)	(5)
All Entrepreneurs	−1.53%	−0.62%	+0.59%	−0.68%	−1.50%
Workers	−0.90%	−0.28%	+0.80%	−0.33%	−1.48%
By Size					
Bottom tercile	−2.10%	−1.00%	+1.05%	−1.35%	−2.55%
Middle tercile	−0.88%	−0.40%	+0.68%	−0.42%	−1.15%
Top tercile	+0.45%	+0.48%	−0.18%	+0.55%	+0.20%
By Leverage					
Bottom tercile	−0.48%	−0.22%	+0.62%	−0.25%	−0.60%
Middle tercile	−1.15%	−0.55%	+0.20%	−0.52%	−1.40%
Top tercile	−1.75%	−0.95%	+0.25%	−1.10%	−2.05%
Effects by size and leverage					
Small, low leverage	−1.50%	−0.72%	+0.18%	−0.80%	−1.85%
Large, low leverage	+0.74%	+0.85%	+0.10%	+0.95%	+0.40%
Small, high leverage	−2.75%	−1.35%	+1.15%	−1.70%	−3.20%
Large, high leverage	−0.65%	−0.30%	+0.25%	−0.58%	−0.85%

Notes: Welfare is measured as the consumption equivalent variation relative to the baseline calibration (1999–2000, pre-$\underline{w}\uparrow$). Columns (1)–(4) represent counterfactual experiments. Column (2) moderates the wage floor increase to 10%. Column (3) removes the wage floor but maintains monopsony power. Column (4) increases the wage floor by 20% while loosening collateral constraints ($\hat{\chi}=2\chi$). Column (5) raises the wage floor by 20% and lowers the degree of home bias ($\hat{\omega}=0.6<0.7$). Size is defined using the log of real total assets; leverage is the ratio of debt to assets. Categories for size–leverage interactions follow median splits.

or size category), I abstract from compositional changes and base the computation on the firms' classification in the "new" regime (e.g., after the 20% wage floor increase). For example, when computing the welfare measure for firms in the bottom tercile of leverage, I first define the indicator

$$\mathbf{1}_{\overline{lev}_1}(\tau,z)=\begin{cases}0, & \text{if } lev^{\underline{w}\uparrow}(\tau,z)\leq\overline{lev}_1\\ 1, & \text{otherwise}\end{cases},$$

where $\overline{lev}_1$ is the cutoff for the first tercile of leverage, based upon the distribution in the post wage floor increase regime. The compensating differential for a firm in the bottom tercile of leverage is then

$$G_{\overline{lev}_1} = \left[\frac{\sum_z \sum_\tau V^{\underline{w}\uparrow}(\tau, z) \mathbf{1}_{\overline{lev}_1}(\tau, z) \phi(\tau, z) / \sum_z \sum_\tau \mathbf{1}_{\overline{lev}_1}(\tau, z) \phi(\tau, z)}{\sum_z \sum_\tau V^{\text{base}}(\tau, z) \mathbf{1}_{\overline{lev}_1}(\tau, z) \phi(\tau, z) / \sum_z \sum_\tau \mathbf{1}_{\overline{lev}_1}(\tau, z) \phi(\tau, z)} \right]^{\frac{1}{1-\gamma}} - 1.$$

A similar procedure is followed for the other tercile of leverage, the tercile of size (based on the log of real assets), and for categories based on the interaction of size and leverage.

Table 10 column (1) shows the welfare measures for the 20 percent wage floor increase. The contractual wage floor increase reduces entrepreneurs' welfare by 1.53 percent. Workers are also harmed, with welfare falling by 0.90 percent; despite earning higher wages under the new regime, workers suffer a loss in consumption due to cuts in employment and price increases. Consistent with the most financially constrained firms being the most affected by the contractual wage floor increase, small, highly levered entrepreneurs face the largest losses, with welfare contracting by 2.75 percent. Large, low-leverage entrepreneurs benefit by gaining market share under the new regime and see modest welfare gains of 0.74 percent. However, the small gains of large firms are insufficient to compensate for the losses incurred by other entrepreneurs or workers.

5.4 Counterfactual analysis

I now consider several counterfactual experiments: A more moderate (10 percent) increase in the contractual wage floor, removing the wage floor ($\underline{w} = 0$), lowering financial frictions $\hat{\chi} = 2\chi$ so that firms are less constrained during the 20 percent increase, and increasing openness to trade $\hat{\omega} = 0.6$. In all cases, results are presented as a percent change from the baseline calibration.

Moderate increase in contractual wage floors: I first conduct a counterfactual exercise to capture the outcome if Italy experienced a more moderate increase in labor costs, with the contractual wage floor increasing by 10 percent rather than the observed 20 percent. The aggregate results for this experiment are reported in column (2) of Table 7, and the welfare results are reported in column (2) of Table 10. Aggregate employment (-3.11%), investment (-2.22), and sales (-1.04) for entrepreneurs all decline relative to the initial stationary equilibrium, but the contractions are muted by roughly half relative to a 20 percent wage floor increase. Consumption still declines for both the worker (-0.56%) and the entrepreneur (-0.97%) under the more moderate increase, as reflected in negative welfare measures.

Additionally, distributional patterns remain intact, with small, high-leverage firms

bearing the brunt of the contraction (-1.35% welfare) relative to large, low-leverage entrepreneurs (+0.85% welfare). While the negative effects are muted for most entrepreneurs, large and low-leverage entrepreneurs see a larger welfare increase under the 10 percent wage floor than under the 20 percent wage floor. This reflects that fewer large, low-leverage firms are constrained in this scenario and thus are better able to take advantage of an increase in market share.

No contractual wage floor: In this experiment, I remove the contractual wage floor, setting $\underline{w}$ to zero. While the wage floor no longer accounts for inefficiencies, labor market imperfections remain: entrepreneurs maintain (now unchecked) monopsony power. The results for this experiment are reported in column (3) of Table 7 and Table 10. Aggregate employment (+1.15%), investment (+1.05%), and sales (+0.62%) for entrepreneurs all increase relative to the baseline 1999 stationary equilibrium. GDP increases (+0.82%), and workers (+1.60%) and entrepreneurs (+0.91%) both consume more. In this experiment, small firms benefit substantially (+1.05% welfare) from eliminating binding wage constraints. In contrast, large firms suffer small welfare losses (-0.18%) because they lose the competitive advantage that wage floors provided by constraining smaller competitors. Workers see relatively small gains (+0.75% welfare) because, while employment increases and prices decrease relative to the baseline, wages fall under unchecked monopsony power. The positive effects here illustrate the inefficiencies of Italian wage floors even before the 20% increase,[21] with aggregate gains arising under unchecked monopsony power.

Reduced financial frictions: Column (4) of Table 7 and Table 10 reports the results of increasing the contractual wage floor by 20% while lessening financial frictions. I reduce financial frictions by doubling the fraction of capital entrepreneurs can use as collateral, $\hat{\chi} = 2\chi$. The 20% wage floor increase still results in substantial damages relative to the baseline economy, with investment (-2.54%), employment (-3.83%), sales (-1.15%), real GDP (-1.35%), and consumption (-0.64% for workers and -1.05% for entrepreneurs) all falling. However, relative to the results of the main experiment in column (1), reducing financial frictions helps to lessen the overall blow of the wage floor increase: The negative welfare impact of the 20 percent increase is cut roughly in half, with entrepreneurs taking a 1.33% welfare loss in the baseline compared to a 0.68% loss under the case with reduced financial frictions. Workers also benefit with welfare losses falling from 0.90% to 0.33%.

Small firms remain significantly disadvantaged (-1.35% welfare) because, while financial constraints are reduced, wage floors and collateral constraints still bind more

[21] This is confirmed in Appendix C.2 when I compute the optimal level of the wage floor.

severely for smaller entities. Large firms continue to benefit (+0.55% in welfare) as improved access to capital amplifies their existing advantages. The welfare improvements from reduced financial frictions are substantial but insufficient to overcome the fundamental size and leverage gradients created by the wage floor increase.

Increased openness: Column (5) of Tables 7 and 10 report the results of the 20 percent wage floor increase in a more open economy. For this experiment, I decrease the home bias parameter from $\omega = 0.7$ to $\hat{\omega} = 0.6$. The results reveal that greater trade openness amplifies the damage from rigid wage floors: Real GDP (-2.96%), aggregate investment (-5.93%), and employment (-7.82%), domestic sales (-2.45), and exports (-4.82%) all contract beyond the baseline effects of the 20 percent wage floor increase reported in column (1). Import competition intensifies competitive pressure on domestic firms. With lower home bias, entrepreneurs face greater substitution toward foreign goods when their costs rise. Additionally, export competitiveness deteriorates. Higher labor costs make Italian goods less competitive in international markets, and greater trade integration means this export channel represents a larger share of potential demand that firms lose access to. Small, highly-leveraged firms experience a 3.20% decline in welfare—substantially worse than the 2.75% loss under baseline openness. These firms face a double squeeze: Higher wages erode their financial position while import competition limits their ability to maintain margins. The price level increase is dampened with greater openness (+1.09% versus +1.34% in the baseline), as import competition imposes some constraint on pass-through. However, this is outweighed by the increased employment losses, as workers' welfare still declines more severely (-1.48% versus -0.90%) in the more open economy.

Italy's integration into the euro area market increased its trade openness precisely as wage rigidities became more binding without a corresponding improvement in labor market flexibility. The model suggests this timing was particularly unfortunate: Rigid wage floors become substantially more costly in open economies where firms face greater competitive pressure and cannot easily pass through cost increases. The interaction effect between trade openness and labor market rigidity accounts for an additional 0.22 percentage points of GDP loss, or about a 12% amplification of the baseline effect.

6 Conclusion

This paper examines how rising unit labor costs, tied to nationally bargained contractual wage floors, affected Italian manufacturing firms and contributed to sectoral stagnation in the two decades following the adoption of the euro. By combining comprehensive

firm-level data with a general equilibrium model of heterogeneous firms featuring wage floors and financial frictions, I provide new insights into the mechanisms underlying Italy's manufacturing decline and the broader interactions between labor market institutions and financial constraints.

The empirical and quantitative analyses support a negative relationship between growth in contractual wage floors and firm-level employment, investment, and sales, with small and highly leveraged firms experiencing dramatically larger contractions in investment and employment. The theoretical framework demonstrates how these firm-level responses aggregate to generate persistent economic effects. The calibrated model shows that Italy's 20 percent wage floor increase reduced aggregate manufacturing investment by 4.8 percent, employment by 6.1 percent, and real GDP by 5.3 percent. Additionally, I find that financial frictions amplified the negative effects of wage rigidity on investment by approximately 30 percent through an interaction effect: Wage floor increases reduce profits, slow balance sheet recovery, and trap firms in low-investment equilibria where they cannot escape collateral constraints.

These findings have several implications for economic policy. First, the results suggest that labor market reforms focused solely on wage moderation may be insufficient to restore competitiveness in Italy due to pervasive financial frictions. Policies that improve access to credit or strengthen firm balance sheets may be equally important for ensuring that labor market flexibility translates into improved firm performance. Second, the results highlight the dangers of implementing rigid wage-setting institutions in environments where firms face significant financing constraints. Additionally, the findings shed light on Italy's broader economic challenges within the euro area. The analysis suggests that Italy's manufacturing decline was not simply the result of an overvalued exchange rate or excessive wage growth, but rather reflected deeper structural problems in the interaction between labor market institutions and the development of the financial system.

Two key limitations are worth underscoring. First, the quantitative results compare stationary equilibria; transitional dynamics—how quickly firms adjust and who exits—are left for future work. Second, the paper abstracts from entry/exit and product-market competition beyond the price level, which may matter for pass-through and markups.

References

Adamopoulou, Effrosyni and Ernesto Villanueva, "Wage determination and the bite of collective contracts in Italy and Spain," *Labour Economics*, 2022, *76*, 102147.

Ahlfeldt, Gabriel, Duncan Roth, and Tobias Seidel, "Optimal minimum wages," CEPR Discussion Papers 16913, C.E.P.R. Discussion Papers 2022.

Alesina, Alberto, Silvia Ardagna, and Vincenzo Galasso, "The Euro and Structural Reforms," NBER Working Papers 14479, National Bureau of Economic Research, Inc 2008.

Amatori, Franco, Matteo Bugamelli, and Andrea Colli, "455 Technology, Firm Size, and Entrepreneurship," in "The Oxford Handbook of the Italian Economy Since Unification," Oxford University Press, 03 2013.

Andreasen, Eugenia, Sofía Bauducco, and Evangelina Dardati, "Capital controls and firm performance," *Journal of International Economics*, 2024, *150*, 103897.

_ , _ , _ , and Enrique G Mendoza, "Beware the Side Effects: Capital Controls, Trade, Misallocation and Welfare," Working Paper 30963, National Bureau of Economic Research February 2023.

Asriyan, Vladimir, Luc Laeven, Alberto Martin, Alejandro Van der Ghote, and Victoria Vanasco, "Falling interest rates and credit reallocation: Lessons from general equilibrium," Economics Working Papers, Department of Economics and Business, Universitat Pompeu Fabra 2022.

Autor, David H., William R. Kerr, and Adriana D. Kugler, "Does Employment Protection Reduce Productivity? Evidence from US States," *The Economic Journal*, 2007, *117* (521), F189–F217.

Bandiera, Oriana, Luigi Guiso, Andrea Prat, and Raffaella Sadun, "Matching Firms, Managers, and Incentives," *Journal of Labor Economics*, 2015, *33* (3), 623–681.

Bello, Salvatore Lo and Lorenzo Pesaresi, "The Concentration Channel of the Minimum Wage," Working Paper March 2025.

Berger, David, Kyle Herkenhoff, and Simon Mongey, "Labor Market Power," *American Economic Review*, April 2022, *112* (4), 1147–93.

_ , _ , and _ , "Minimum Wages, Efficiency, and Welfare," *Econometrica*, 2025, *93* (1), 265–301.

Boeri, Tito, Andrea Ichino, Enrico Moretti, and Johanna Posch, "Wage Equalization and Regional Misallocation: Evidence from Italian and German Provinces," *Journal of the European Economic Association*, 07 2021, *19* (6), 3249–3292.

Bofinger, Peter, "German wage moderation and the EZ Crisis," VOXEU Column 1431, CEPR Nov 2015.

Brandolini, Andrea, Piero Casadio, Piero Cipollone, Marco Magnani, Alfonso Rosolia, and Roberto Torrini, *Employment Growth in Italy in the 1990s: Institutional Arrangements and Market Forces*, Heidelberg: Physica-Verlag HD,

Buera, Francisco J. and Benjamin Moll, "Aggregate Implications of a Credit Crunch: The Importance of Heterogeneity," *American Economic Journal: Macroeconomics*, July 2015, *7* (3), 1–42.

_ , Joseph P. Kaboski, and Yongseok Shin, "Finance and Development: A Tale of Two Sectors," *American Economic Review*, August 2011, *101* (5), 1964–2002.

Bugamelli, Matteo, Francesca Lotti, Monica Amici, Emanuela Ciapanna, Fabrizio Colonna, Francesco D'Amuri, Silvia Giacomelli, Andrea Linarello, Francesco Manaresi, Giuliana Palumbo, Filippo Scoccianti, and Enrico Sette, "Productivity growth in Italy: a tale of a slow-motion change," Questioni di Economia e Finanza (Occasional Papers) 422, Bank of Italy, Economic Research and International Relations Area 2018.

Calcagnini, Giorgio, Germana Giombini, and Enrico Saltari, "Financial and labor market imperfections and investment," *Economics Letters*, 2009, *102* (1), 22–26.

Cingano, Federico, Marco Leonardi, Julián Messina, Giovanni Pica, Luigi Pistaferri, and Etienne Wasmer, "The effects of employment protection legislation and financial market imperfections on investment: evidence from a firm-level panel of EU countries [with Discussion]," *Economic Policy*, 2010, *25* (61), 117–163.

Şebnem Kalemli-Özcan, Bent E. Sørensen, Carolina Villegas-Sanchez, Vadym Volosovych, and Sevcan Yeşiltaş, "How to Construct Nationally Representative Firm-Level Data from the Orbis Global Database: New Facts on SMEs and Aggregate Implications for Industry Concentration," *American Economic Journal: Macroeconomics*, April 2024, *16* (2), 353–74.

D'Amuri, Francesco and Cristina Giorgiantonio, "The Institutional and Economic Limits to Bargaining Decentralization in Italy," IZA Policy Papers 98, Institute of Labor Economics (IZA) 2015.

Daruich, Diego, Sabrina Di Addario, and Raffaele Saggio, "The Effects of Partial Employment Protection Reforms: Evidence from Italy," *The Review of Economic Studies*, 02 2023, *90* (6), 2880–2942.

Daveri, Francesco, Guido Tabellini, Samuel Bentolila, and Harry Huizinga, "Unemployment, Growth and Taxation in Industrial Countries," *Economic Policy*, 2000, *15* (30), 49–104.

Destefanis, Sergio, Giuseppe Mastromatteo, and Giovanni Verga, "WAGES AND MONETARY POLICY IN ITALY BEFORE AND AFTER THE WAGE AGREEMENTS," *Rivista Internazionale di Scienze Sociali*, 2005, *113* (2), 289–318.

Devicienti, Francesco, Agata Maida, and Paolo Sestito, "Downward Wage Rigidity in Italy: Micro-Based Measures and Implications," *The Economic Journal*, 2007, *117* (524), F530–F552.

_ **and Bernardo Fanfani**, "Firms' margins of adjustment to wage growth: the case of Italian collective bargaining," *Economica*, 2025, *92* (365), 107–149.

_ **,** _ **, and Agata Maida**, "Collective Bargaining and the Evolution of Wage Inequality in Italy," *British Journal of Industrial Relations*, 2019, *57* (2), 377–407.

Diéz-Catalán, Luis and Ernesto Villanueva, "Contract staggering and unemployment during the great recession: evidence from Spain," Working Papers 1431, Banco de España Jan 2015.

Drechsel-Grau, Moritz, "Employment and Reallocation Effects of Higher Minimum Wages," CESifo Working Paper Series 10412, CESifo 2023.

Engbom, Niklas and Christian Moser, "Earnings Inequality and the Minimum Wage: Evidence from Brazil," *American Economic Review*, December 2022, *112* (12), 3803–47.

Garcia-Macia, Daniel, "Labor Costs and Corporate Investment in Italy," IMF Working Papers 2020/038, International Monetary Fund 2020.

Gopinath, Gita, Şebnem Kalemli-Özcan, Loukas Karabarbounis, and Carolina Villegas-Sanchez, "Capital Allocation and Productivity in South Europe*," *The Quarterly Journal of Economics*, 06 2017, *132* (4), 1915–1967.

Hall, Peter A., "Varieties of Capitalism and the Euro Crisis," *West European Politics*, 2014, *37* (6), 1223–1243.

Hancké, Bob, *Unions, Central Banks, and EMU: Labour Market Institutions and Monetary Integration in Europe*, Oxford University Press, 04 2013.

Haucap, Justus, Uwe Pauly, and Christian Wey, "Collective wage setting when wages are generally binding An antitrust perspective," *International Review of Law and Economics*, 2001, *21* (3), 287–307.

Hijzen, Alexander and Pedro S. Martins, "No Extension without Representation? Evidence from a Natural Experiment in Collective Bargaining," Working Papers 68, Queen Mary, University of London, School of Business and Management, Centre for Globalisation Research Jul 2016.

Hopenhayn, Hugo and Richard Rogerson, "Job Turnover and Policy Evaluation: A General Equilibrium Analysis," *Journal of Political Economy*, 1993, *101* (5), 915–938.

Hurst, Erik, Patrick J Kehoe, Elena Pastorino, and Thomas Winberry, "The Distributional Impact of the Minimum Wage in the Short and Long Run," Working Paper 30294, National Bureau of Economic Research July 2022.

IStat, "La flessibilita del mercato del lavoro nel periodo 1995-96," Collana Informazioni., Istat 2000.

Kangur, Alvar, "Competitiveness and Wage Bargaining Reform in Italy," IMF Working Papers 2018/061, International Monetary Fund 2018.

Karabarbounis, Loukas, Jeremy Lise, and Anusha Nath, "Minimum Wages and Labor Markets in the Twin Cities," Working Paper 30239, National Bureau of Economic Research July 2022.

Leonardi, Marco, Michele Pellizzari, and Domenico Tabasso, "Wage Compression within the Firm: Evidence from an Indexation Scheme," *The Economic Journal*, 05 2019, *129* (624), 3256–3291.

Magruder, Jeremy R., "High Unemployment Yet Few Small Firms: The Role of Centralized Bargaining in South Africa," *American Economic Journal: Applied Economics*, July 2012, *4* (3), 138–66.

Manacorda, Marco, "Can the Scala Mobile Explain the Fall and Rise of Earnings Inequality in Italy? A Semiparametric Analysis, 1977–1993," *Journal of Labor Economics*, 2004, *22* (3), 585–613.

Martins, Pedro S., "30,000 Minimum Wages: The Economic Effects of Collective Bargaining Extensions," *British Journal of Industrial Relations*, 2021, *59* (2), 335–369.

Midrigan, Virgiliu and Daniel Yi Xu, "Finance and Misallocation: Evidence from Plant-Level Data," *American Economic Review*, February 2014, *104* (2), 422–58.

Pellegrino, Bruno and Luigi Zingales, "Diagnosing the Italian Disease," Working Paper 23964, National Bureau of Economic Research October 2017.

Sokolova, Anna and Todd Sorensen, "Monopsony in Labor Markets: A Meta-Analysis," *ILR Review*, 2021, *74* (1), 27–55.

Tokarski, Pawel, "Divergence and diversity in the Euro Area: The case of Germany, France and Italy," SWP Research Paper 6/2019, Stiftung Wissenschaft und Politik, Berlin 2019.

A Details of Empirical Analysis

A.1 Data

Table A.1: Summary statistics of key variables

	Mean	Standard deviation
Value Added	2.98	6.91
Employment	35.99	71.5
Wage Bill	1.75	3.01
Capital Stock	2.58	4.07
Total Assets	9.5	15.01
Total Liabilities	6.35	9.75
Age	22.6	14.2
Sales	9.2	10.1
Cash flow ratio	0.07	0.14
Leverage	0.19	0.21
Interest exp. ratio	0.18	0.75
Labor share	0.67	0.28

Notes: Summary statistics for key variables from Orbis data. The interest expense ratio is the ratio of interest expense to EBITDA; leverage is the ratio of total debt to total assets; the cash flow ratio is cash flow to total assets.

Table A.1 presents summary statistics for key variables in the firm-level dataset. Aside from employment, age, and ratio variables, all averages are expressed in millions of euros. Value added, wage bill, total assets, and total liabilities are deflated with value added price indices at the two digits industry level with a base year of 2015. The capital stock is the sum of tangible and intangible fixed assets and is deflated at the NACE-2 level with the price of investment goods with a base year of 2015. The age of the firm is the difference between the year of the balance sheet information and the year of incorporation of the firm plus one. Means and standard deviations are computed for the period 1999-2019.

Table A.2 presents coverage statistics for employment, wage bill, and gross output in the Orbis sample relative to official statistics reported in Eurostat's SBS for the Italian manufacturing sector. Coverage is high throughout the sample for all three variables, reaching close to 90% in the later years.

Table A.3 reports the share of economic activity accounted for by firms across to different size categories in 2010 in the Orbis sample and in Eurostat SBS for Italy's manufacturing sector. The sample is fairly representative in its ability to capture share of

Table A.2: Coverage in Orbis relative to Eurostat SBS

Year	Employment	Wage bill	Gross Output
1999	0.45	0.59	0.51
2000	0.55	0.63	0.57
2001	0.68	0.62	0.65
2002	0.58	0.69	0.71
2003	0.63	0.68	0.70
2004	0.59	0.71	0.73
2005	0.65	0.72	0.77
2006	0.62	0.73	0.79
2007	0.66	0.73	0.79
2008	0.62	0.84	0.90
2009	0.70	0.81	0.86
2010	0.73	0.83	0.87
2011	0.74	0.86	0.89
2012	0.77	0.85	0.86
2013	0.75	0.81	0.79
2014	0.84	0.88	0.87
2015	0.86	0.89	0.89
2016	0.85	0.87	0.86
2017	0.81	0.88	0.85
2018	0.83	0.89	0.87
2019	0.87	0.88	0.89

Notes: Manufacturing sector coverage statistics for the wage bill and value added relative to statistics reported in Eurostat SBS for Italy.

Table A.3: Sample representativeness by size (2010)

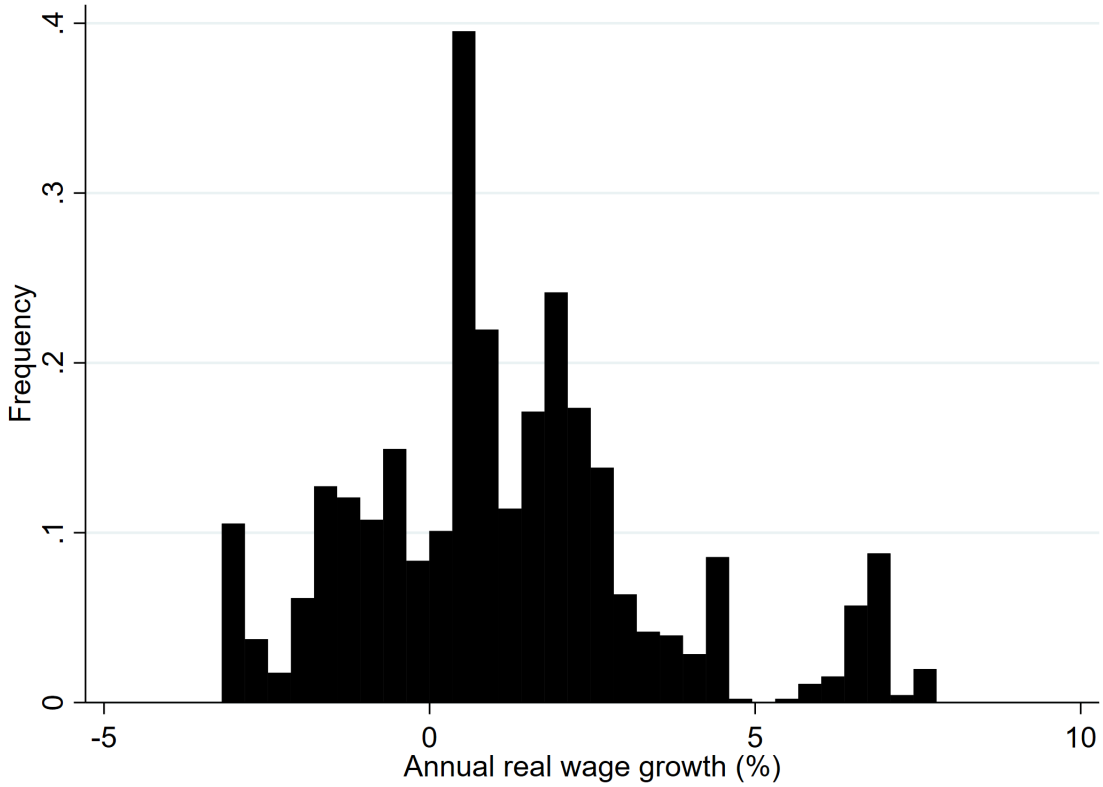
Source	Size	Employment	Wage bill	Gross Output
Historical Orbis	1-19 employees	0.15	0.12	0.13
	20-249 employees	0.55	0.52	0.52
	250+ employees	0.30	0.36	0.35
Eurostat (SBS)	1-19 employees	0.38	0.21	0.20
	20-249 employees	0.38	0.43	0.42
	250+ employees	0.24	0.36	0.38

Notes: Share of economic activity accounted for by firms of different size categories in 2010, Orbis vs. Eurostat SBS.

economic activity accounted for by small and medium firms.

Figure A.1 provides a histogram for annual real wage floor growth across sub-sectors from 2005 to 2019. The average annual growth rate over the sample is around 1 percent. There is substantial variation across sub-sectors and years: Annual changes in contractual wage floors range from –3 to +8 percent across industries and years.

Figure A.1: Real wage growth by sector-year, 2005-2019



Annual real wage growth is measured at the NACE-3 level for years 2005-2019. Wage floor indices are deflated using the NACE-2 level value added deflators. Sources: IStat, Eurostat

A.2 Empirical results

Table A.4 reports the full results for the main regression specification in Equation 1. Full controls include the lagged labor share α_{ijt-1}, The lagged dependent variable $\%\Delta Y_{it-1}$, the lagged debt service ratio computed as the ratio of financial expenses to EBITDA DSR_{ijt-1}, lagged leverage lev_{ijt-1}, lagged cashflow ratio computed as cashflow to total assets CF_{ijt-1}, and indicators for tercile of size based on the log of lagged real total assets $size_2$ and $size_3$. Results are reported for the preferred specification with firm and sector-year fixed effects (column 1 for each dependent variable), for a specification where sector-year fixed effects are replaced by year fixed effects (column 2 for each dependent variable), and for specification with out the full set of firm controls (column 3). Across all specifications and all dependent variables, the regression predicts a negative relationship between contractual wage floor growth and firm outcomes.

Table A.4: Wage floor growth and firm-level outcomes, full results

	Capital Growth			Labor Growth			Sales Growth		
Variables	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
$\alpha_{it-1} \times \%\Delta\bar{w}_{jt}$	-0.399***	-0.279**	-0.191*	-0.320***	-0.252**	-0.232**	-0.275**	-0.366**	-0.141
	(0.126)	(0.089)	(0.111)	(0.163)	(0.108)	(0.105)	(0.140)	(0.143)	(0.140)
α_{it-1}	0.081***	0.077***	0.064**	-0.229***	-0.228***	-0.237***	0.240***	0.253***	0.219***
	(0.005)	(0.005)	(0.005)	(0.007)	(0.007)	(0.006)	(0.007)	(0.007)	(0.007)
$\%\Delta Y_{it-1}$	-0.083***	-0.082***		-0.254***	-0.253***		-0.191***	-0.186***	
	(0.003)	(0.003)		(0.004)	(0.004)		(0.005)	(0.005)	
DSR_{it-1}	-0.002	-0.000		-0.003	-0.002		-0.005	-0.000	
	(0.004)	(0.004)		(0.004)	(0.004)		(0.004)	(0.005)	
lev_{it-1}	0.081***	0.082***		-0.034***	-0.035***		-0.060***	-0.062***	
	(0.005)	(0.005)		(0.006)	(0.006)		(0.004)	(0.004)	
CF_{it-1}	0.210***	0.213***		-0.150***	-0.140***		-0.205***	-0.201***	
	(0.021)	(0.021)		(0.011)	(0.012)		(0.019)	(0.020)	
size_2	0.049***	0.048***		0.016***	0.016***		0.056***	0.057***	
	(0.003)	(0.003)		(0.003)	(0.003)		(0.003)	(0.003)	
size_3	0.163***	0.162***		0.033***	0.034***		0.165***	0.169***	
	(0.005)	(0.005)		(0.005)	(0.005)		(0.005)	(0.005)	
Observations	405,092	405,249	405,092	339,666	339,867	353,674	405,061	405,203	405,061
Adj R^2	0.342	0.351	0.342	0.123	0.121	0.115	0.455	0.439	0.397
Firm Controls	YES	YES	NO	YES	YES	NO	YES	YES	NO
Firm FE	YES	YES	YES	YES	YES	YES	YES	YES	YES
Sector-year FE	YES	NO	YES	YES	NO	YES	YES	NO	YES
Year FE	NO	YES	NO	NO	YES	NO	NO	YES	NO

Notes: Full results for main regression specified in Equation 1. $\%\Delta Y_{it-1}$ is the lagged dependent variable, DSR is the firm's debt service ratio, lev is the firm's leverage, CF is the entrepreneur's cash flow ratio. $size_x$ are indicators for the tercile of firm size (log of lagged real total assets). Standard errors in parentheses clustered at the sector-year level. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

B Model Details

B.1 Marginal revenue products

The value of its sales defines an entrepreneur's revenue:

$$Rev_i = \varphi_i q_i.$$

The marginal revenue products of labor and capital are then given by

$$MRPN_i \equiv \delta Rev_i/\delta n_i \quad \text{and} \quad MRPK_i \equiv \delta Rev_i/\delta k_i.$$

To get the expressions for $MRPK_i$ and $MRPN_i$ used in Equations 13 and 14, start by taking derivatives of Rev_i with respect to n and k:

$$MRPN = [\varphi_i + q_i(\delta\varphi_i/\delta q_i)](\delta q_i/\delta n_i), \tag{22}$$
$$MRPK = [\varphi_i + q_i(\delta\varphi_i/\delta q_i)](\delta q_i/\delta k_i). \tag{23}$$

Optimality requires that the price is

$$\varphi_i = \left[\frac{(P^Q)^\sigma Q}{z_i k_i^\alpha n_i^{1-\alpha}}\right]^{\frac{1}{\sigma}} = \left[\frac{(P^Q)^\sigma Q}{q_i}\right]^{\frac{1}{\sigma}}.$$

Using this,

$$\frac{\delta\varphi_i}{\delta q_i} = -\frac{\varphi_i}{\sigma}\frac{1}{q_i}.$$

Substituting these expressions into Equations 22 and 23 yields

$$MRPN_i = \frac{\varphi_i}{\varsigma}\left(\frac{\delta q_i}{\delta n_i}\right), \qquad MRPK = \frac{\varphi_i}{\varsigma}\left(\frac{\delta q_i}{\delta k_i}\right), \tag{24}$$

where $\varsigma = \sigma/(\sigma-1)$ is the markup charged by the monopolistically competitive entrepreneur.

Differentiating q_i with respect to n_i and k_i and substituting into Equation 24 gives the definitions of $MRPK_i$ and $MRPN_i$ used in presenting the first order conditions from the second stage problem:

$$MRPN = \frac{\varphi_i}{\varsigma} z_i(1-\alpha)\left(\frac{k_i}{n_i}\right)^\alpha, \qquad MRPK = \frac{\varphi_i}{\varsigma} z_\alpha \left(\frac{n_i}{k_i}\right)^{1-\alpha}.$$

B.2 Shadow wage notation

Throughout the appendix, I follow Berger et al. (2025) and adopt the concepts of *shadow wages* and *shadow markdowns* to characterize the solution to the entrepreneur's problem more concisely. This is helpful for aggregation because in the presence of a wage floor, some entrepreneurs' wages are not optimal (Region II), and some entrepreneurs' wages are not allocative due to the deviation from the labor supply curve (Region III). Using the normalized multiplier on the rationing constraint, ν_i, from the worker's problem, define the shadow (real) wage $\tilde{w}_i$ and the shadow markdown $\tilde{\mu}_i$ as:

$$\tilde{w}_i \equiv \nu_i \frac{W_i}{P} = \left(\frac{n_i}{N}\right)^{\frac{1}{\theta}} \left(\frac{N}{\bar{\varsigma}}\right)^{\frac{1}{\varsigma}}, \quad \tilde{\mu}_i \equiv \frac{\tilde{w}_i}{MRPL_i}.$$

The shadow wage captures two ideas: first, it is the relevant allocative price for household employment, as it always places the household on its labor supply curve. Second, because $\tilde{w}_i = \nu_i W_i/P \leq W_i/P$, $\tilde{w}_i$ encodes the degree to which the rationing constraint binds under a given wage floor. Since shadow wages determine quantities, they are useful for aggregation.

Now define $\tilde{W} \equiv \left(\int \tilde{w}_i^{1+\theta} di\right)^{\frac{1}{1+\theta}}$. Using this, I can write entrepreneur-level and aggregate labor supply curves as

$$n_i = \left(\frac{\tilde{w}_i}{\tilde{W}}\right)^{\theta} N \quad \text{and} \quad N_t = \bar{\varsigma}\left(\tilde{W}\right)^{\varsigma}.$$

Shadow markdowns capture inefficiencies arising from market power in Region I, diminished market power in Region II, and binding rationing constraints due to a wage floor in Region III.

B.3 Static effects without financial frictions

Consider the static effects of a wage floor in an environment with no financial frictions. To remove financial frictions from the model, assume $\chi \to \infty$ so that the collateral constraint never binds. In this case, all entrepreneurs equate their marginal revenue products of capital:

$$MRPK_i = P'(i^* + \delta) \quad \forall i.$$

In the absence of financial frictions, there is no dispersion in $MRPK$ across entrepreneurs: If an entrepreneur's initial capital allocation, $\underline{k}(z_i)$, is below its steady state value $k^*(z_i)$, the

entrepreneur will borrow however much is needed to jump to its optimal scale immediately. Thus, in solving the second-stage problem, net worth becomes irrelevant; an entrepreneur's allocations of k_i and n_i and its region will depend only on z_i and the prevailing wage floor $\underline{w}$.

For entrepreneurs in region I, optimal capital and labor allocations in a stationary equilibrium without financial frictions are

$$k_I^*(z) = \left(\frac{\varphi Q^{\frac{1}{\sigma}}}{P\vartheta} z^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma(1+\theta)}{\alpha+\theta+\sigma-\alpha\sigma}} \left(\frac{\alpha}{(i+\delta)}\right)^{\frac{\sigma-\alpha\theta+\theta+\sigma\alpha\theta}{\alpha+\theta+\sigma-\alpha\sigma}} \left(\frac{\theta}{\theta+1}\frac{(1-\alpha)}{\tilde{W}} N^{\frac{1}{\theta}}\right)^{\frac{(1-\alpha)(\sigma-1)\theta}{\alpha+\theta+\sigma-\alpha\sigma}}, \quad (25)$$

and

$$n_I^*(z) = \left(\frac{1-\alpha}{\vartheta}\frac{\theta}{\theta+1}\frac{\varphi Q^{\frac{1}{\sigma}}}{P}\frac{N^{\frac{1}{\theta}}}{\tilde{W}}(z k_I^*(z)^{\alpha})^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma\theta}{\sigma-\alpha\theta+\theta+\sigma\alpha\theta}}. \quad (26)$$

For entrepreneurs in region II,

$$k_{II}^*(z,\underline{w}) = \left(\frac{\varphi Q^{\frac{1}{\sigma}}}{\vartheta}\frac{\alpha}{P(i^*+\delta)} z^{\frac{\sigma-1}{\sigma}} \left(\left(\frac{\underline{w}}{\tilde{W}}\right)^{\theta} N\right)^{\frac{(\sigma-1)(1-\alpha)}{\sigma}}\right)^{\frac{\sigma}{\sigma+\alpha-\alpha\sigma}}, \quad (27)$$

and

$$n_{II}^*(\underline{w}) = \left(\frac{\underline{w}}{\tilde{W}}\right)^{\theta} N. \quad (28)$$

For entrepreneurs in region III,

$$k_{III}^*(z,\underline{w}) = \left(\frac{\varphi Q^{\frac{1}{\sigma}}}{P\vartheta} z^{\frac{\sigma-1}{\sigma}}\right)^{\sigma} \left(\frac{\alpha}{(i^*+\delta)}\right)^{\sigma\alpha+1-\alpha} \left(\frac{1-\alpha}{\underline{w}}\right)^{(1-\alpha)(\sigma-1)}, \quad (29)$$

and

$$n_{III}^*(z,\underline{w}) = \left(\frac{\varphi Q^{\frac{1}{\sigma}}}{\vartheta}\frac{1-\alpha}{P\underline{w}}(z k_{III}^*(z)^{\alpha})^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\alpha\sigma+1-\alpha}}. \quad (30)$$

Thus, to determine an entrepreneur's region of constraint under a given $\underline{w}$, compute the shadow markdown for an entrepreneur, assuming it falls into region II,

$$\tilde{\mu}^{(2)}(z_i) = P'\bar{w}/MRPL(z_i, k_{II}^*(z_i), n_{II}^*(z_i)),$$

and use this to determine an entrepreneur's region, r, as

$$r(z_i) = \begin{cases} I & \text{if} \quad \tilde{\mu}^{(2)}(z_i) < \frac{\theta}{1+\theta} \\ II & \text{if} \quad \frac{\theta}{1+\theta} \leq \tilde{\mu}^{(2)}(z_i) \geq 1 \\ III & \text{if} \quad \tilde{\mu}^{(2)}(z_i) > 1. \end{cases} \tag{31}$$

Proposition B.3.1: Comparing static effects (holding aggregates constant), when the wage floor is increased from $\underline{w}_1$ to $\underline{w}_2 > \underline{w}_1$ such that entrepreneur i moves from region I to region II in an environment with no collateral constraint ($\chi \to \infty$), the entrepreneur will increase allocations of k and n, decrease prices (φ) and increase the quantity produced q.

Proof. For the lowest possible wage floor $\underline{w}_L$ such that an entrepreneur enters region II, the markdown is $\mu_i^{II}(\underline{w}_L, z_i) = \frac{\theta}{1+\theta}$. While the entrepreneur is still in region I ($\forall \underline{w} < \underline{w}_L$), the markdown is $\mu_i^{I}(z_i) = \frac{\theta}{1+\theta}$. Thus, since the entrepreneur retains the full markdown at the border between region I and region II, the lowest wage floor at which the entrepreneur enters region II must be equal to the wage level the entrepreneur optimally sets in region I. Thus, the labor allocation in region II under $\underline{w}_L$ must be equivalent to that chosen in region I:

$$\underline{w}_L = w^I(z_i) \quad \Rightarrow \quad \left(\frac{n_I^*(z)}{N}\right)^{\frac{1}{\theta}} \tilde{W} = \left(\frac{n_{II}^*(z, \underline{w}_L)}{N}\right)^{\frac{1}{\theta}} \tilde{W} \quad \Rightarrow \quad n_I^*(z) = n_{II}^*(z, \underline{w}_L).$$

In an environment without collateral constraints, the $MRPK$ does not vary with $\underline{w}$ in a partial equilibrium. Thus, $MRPK(z, k_I^*(z), n_I^*(z)) = MRPK(z, k_{II}^*(z, \underline{w}), n_{II}^*(z, \underline{w}))$. Because $n_I^*(z) = n_{II}^*(z, \underline{w}_L)$, this means that $k_I^*(z) = k_{II}^*(z, \underline{w}_L)$.

Next, note that in region II, capital and labor are increasing in $\underline{w}$. From equations 27 and 28,

$$\frac{\partial k_{II}^*(z, \underline{w})}{\partial \underline{w}} > 0 \quad \text{and} \quad \frac{\partial n_{II}^*(z, \underline{w})}{\partial \underline{w}} > 0.$$

Thus, since $\underline{w}_L$ is the lowest wage such that an entrepreneur is in region II, for any $\underline{w}_2$ such that the entrepreneur is in region II,

$$n_{II}^*(z, \underline{w}_2) \geq n_{II}^*(z, \underline{w}_L) = n_I^*(z) \quad \text{and} \quad k_{II}^*(z, \underline{w}_2) \geq k_{II}^*(z, \underline{w}_L) = k_I^*(z).$$

Given $q_i = z_i k_i^{\alpha} n_i^{1-\alpha}$ is increasing in k and n, and $\varphi_i = \left[\frac{(\varphi)^{\sigma} Q}{z_i k_i^{\alpha} n_i^{1-\alpha}}\right]^{\frac{1}{\sigma}}$ is decreasing in k and n, it follows that

$$q_{II}^*(z, \underline{w}_2) \geq q_I^*(z) \quad \text{and} \quad \varphi_{II}^*(z, \underline{w}_2) \leq \varphi_I^*(z).$$

□

Proposition B.3.2: Under $\underline{w}$ such that an entrepreneur is in region III, optimal allocations of capital and labor decrease with any increase in the wage floor.

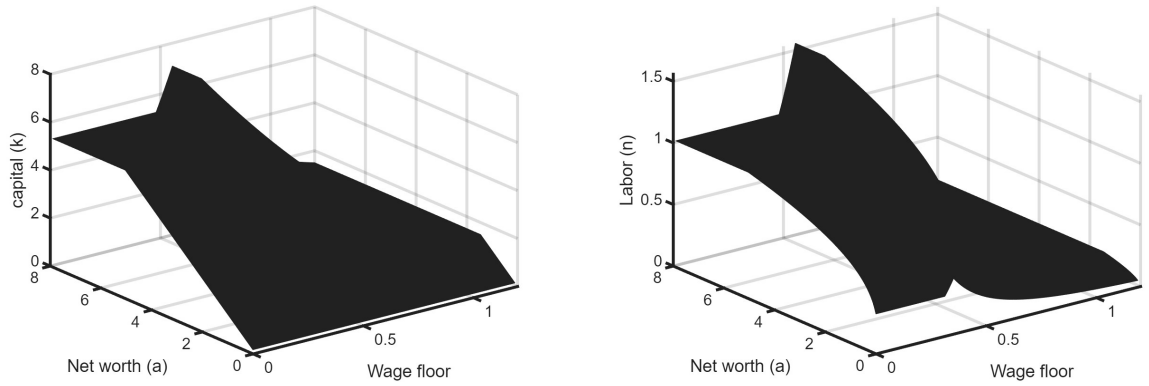
Proof. In the limit such that $\underline{w} \to \infty$, optimal allocations of capital and labor go to zero, and the entrepreneur ceases production.

$$\frac{\partial x^*_{III}(z, \underline{w})}{\partial \underline{w}} < 0 \quad \text{and} \quad \lim_{w \to \infty} x^*_{III}(z, \underline{w}) = 0 \quad x \in \{k, n, q\}.$$

□

B.4 Static effects with financial frictions

Figure B.1: Static choices of capital and labor



(a) Capital (b) Labor

Note: This figure plots the entrepreneur's static choices of capital and labor as a function of productivity, net worth, and the wage floor.

This appendix provides details for the static effects of wage floors in the full environment with collateral constraints ($\chi > 0$).

Proposition B.4.1: For χ sufficiently low such that the collateral constraint binds for entrepreneur i, $MRPK_i > \overline{MRPK}$ and $k_i < \bar{k}_i$, where $\bar{k}_i$ is the entrepreneur's steady state (optimal level) of capital[22] and $\overline{MRPK} = P(i^* + \delta)$.

Proof. From equation 13,

$$MRPK_i = P(i^* + \delta) + \eta(1 - \chi),$$

[22]The steady state level of k_i will depend upon the prevailing wage floor $\underline{w}$ and corresponds to the capital allocations described in the environment with no collateral constraints.

under a binding collateral constraint, $\eta > 0$, so $MRPK_i > \overline{MRPK}$. entrepreneurs with $k_i < \bar{k}_i$, want to borrow as much as possible to reach their optimal scale $\bar{k}_i$. If required debt exceeds $\chi \bar{k}_i$, jumping to the optimal scale is infeasible, and the entrepreneur will set investment as high as the constraint allows:

$$k_i'(a_i') = \frac{a_i'}{(1-\chi)}.$$

Thus, the entrepreneur must accumulate capital gradually as it grows its net worth. □

Assuming the collateral constraint binds, an entrepreneur's investment choice depends only on its net worth, regardless of the wage floor or corresponding region

$$k_I(a) = k_{II}(a) = k_{III}(a) = \frac{a}{1-\chi},$$

and its choice of labor in each region under a given $\underline{w}$ now depends on both net worth and productivity:

$$n_I^*(z,a) = \left(\frac{1-\alpha}{\vartheta} \frac{\theta}{\theta+1} \frac{P^Q Q^{\frac{1}{\sigma}}}{P} \frac{N^{\frac{1}{\theta}}}{\tilde{W}} (z k_I(a)^{\alpha})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma\theta}{\sigma-\alpha\theta+\theta+\sigma\alpha\theta}} \tag{32}$$

$$n_{II}^*(\underline{w}) = \left(\frac{\underline{w}}{\tilde{W}} \right)^{\theta} N. \tag{33}$$

$$n_{III}^*(z,a,\underline{w}) = \left(\frac{P^Q Q^{\frac{1}{\sigma}}}{\vartheta} \frac{1-\alpha}{P\underline{w}} (z k_{III}(a)^{\alpha})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\alpha\sigma+1-\alpha}}. \tag{34}$$

Thus, determining an entrepreneur's region of constraint under a given $\underline{w}$ will now depend jointly on productivity and on net worth. To see this, compute the shadow markdown for an entrepreneur assuming it falls into region II,

$$\tilde{\mu}^{(2)}(z_i, a_i, \underline{w}) = P'\bar{w}/MRPL(z_i, k_{II}(a_i), n_{II}^*(\underline{w})),$$

and use this to determine an entrepreneur's region, r_i, as

$$r_i(z_i, a_i, \underline{w}) = \begin{cases} I & \text{if} \quad \tilde{\mu}^{(2)}(z_i, a, \underline{w}) < \frac{\theta}{1+\theta} \\ II & \text{if} \quad \frac{\theta}{1+\theta} \leq \tilde{\mu}^{(2)}(z_i, a, \underline{w}) \geq 1 \\ III & \text{if} \quad \tilde{\mu}^{(2)}(z_i, a_i, \underline{w}) > 1 \end{cases}. \tag{35}$$

Note that $\tilde{\mu}^{(2)}(z_i, a_i, \underline{w})$ is increasing in $\underline{w}$ and decreasing in a and z. Two entrepreneurs of the same productivity can fall into different regions under a given $\underline{w}$: The entrepreneur with a higher net worth will be less constrained by the wage floor.

Proposition B.4.2: Consider an entrepreneur i in region I under $\underline{w}_1$. Assume the entrepreneur is bound by the collateral constraint. If the wage floor is increased to a level $\underline{w}_2$ such that the entrepreneur moves from region I to region II, there will be no change in the entrepreneur's choice of capital k_i, increases in labor n_i and production q_i, and decreases in prices (φ_i).

Proof. From Proposition 1, in the absence of binding collateral constraints, the entrepreneur would increase its allocation of capital in response to an increase in the wage floor. However, an entrepreneur facing collateral constraints can only set investment as high as the constraint allows. Thus, the entrepreneur's capital choice depends only on net worth and is invariant with respect to the change in wage floor

$$k_I(a_i) = k_{II}(a_i) = \frac{a_i}{1-\chi}.$$

For the lowest possible wage floor $\underline{w}_L$ such that the entrepreneur enters region II, the markdown is $\mu_i^{II}(\underline{w}_L, z_i, a_i) = \frac{\theta}{1+\theta}$. While the entrepreneur is still in region I ($\forall \underline{w} < \underline{w}_L$), the markdown is $\mu_i^{I}(z_i) = \frac{\theta}{1+\theta}$. Thus, since the entrepreneur retains the full markdown at the border between region I and region II, the lowest wage floor at which the entrepreneur enters region II must be equal to the wage level the entrepreneur optimally sets in region I. Thus, the labor allocation in region II under $\underline{w}_L$ must be equivalent to that chosen in region I:

$$\begin{aligned} \underline{w}_L = w^I(z_i, a_i) \quad &\Rightarrow \quad \left(\frac{n_I^*(z_i, a_i)}{N}\right)^{\frac{1}{\theta}} \tilde{W} = \left(\frac{n_{II}^*(z_i, a_i, \underline{w}_L)}{N}\right)^{\frac{1}{\theta}} \tilde{W} \\ &\Rightarrow \quad n_I^*(z_i, a_i) = n_{II}^*(z_i, a_i, \underline{w}_L). \end{aligned}$$

Next, note that in region II, labor is increasing in $\underline{w}$. From equations 34,

$$\frac{\partial n_{II}^*(z_i, a_i, \underline{w})}{\partial \underline{w}} > 0.$$

Thus, since $\underline{w}_L$ is the lowest wage such that an entrepreneur is in region II, for any $\underline{w}_2$ such that the entrepreneur is in region II,

$$n_{II}^*(z, a_i, \underline{w}_2) \geq n_{II}^*(z_i, a_i, \underline{w}_L) = n_I^*(z_i, a_i)$$

Given $q_i = z_i k_i^{\alpha} n_i^{1-\alpha}$ is increasing in k and n, and $\varphi_i = \left[\frac{(\varphi)^{\sigma} Q}{z_i k_i^{\alpha} n_i^{1-\alpha}}\right]^{\frac{1}{\sigma}}$ is decreasing in n, it follows that

$$q_{II}^*(z_i, a_i, \underline{w}_2) \geq q_I^*(z_i, a_i) \quad \text{and} \quad \varphi_{II}^*(z_i, a_i \underline{w}_2) \leq \varphi_I^*(z_i, a_i).$$

□

Proposition B.4.3: A financially constrained entrepreneur is more sensitive to increases in the wage floor than an unconstrained entrepreneur: Comparing two entrepreneurs of the same productivity, an entrepreneur bound by its collateral constraint will move from region I to region II and region II to region III at lower levels of $\underline{w}$. Thus, the scope for increasing investment and employment via the wage floor will be more limited than in an environment with no financial frictions

Proof. Consider two entrepreneurs of the same z, and a wage floor $\underline{w}$. One entrepreneur has enough net worth, a^h, such that collateral constraints do not bind, and the other has a^ℓ such that it is subject to the constraint. Thus, regardless of the wage floor, the high-net-worth entrepreneur will be at its optimal scale, $k^h = k^*$. Since they have the same productivity, the two entrepreneurs have the same optimal scale, but per proposition 3, the low net worth entrepreneur will have

$$k(a^\ell) = \frac{a^\ell}{1-\chi} < k^* = k^h$$

, due to the binding collateral constraint. Note that, assuming the entrepreneurs are both in region II, they would choose the same allocation of labor since the region 2 allocation is pinned down solely by the wage floor and aggregates:

$$n^\ell_{II} = n^h_{II} = \left(\frac{\underline{w}}{\tilde{W}}\right)^\theta N.$$

Thus, since MRPL is increasing in k,

$$MRPL(z, k^h, n^h_{II}) > MRPL(z, k(a^\ell), n^\ell_{II}).$$

Let $\underline{w}^h_I$ and $\underline{w}^\ell_I$ denote the lowest wage floor such that each entrepreneur is in region II, and $\underline{w}^h_{III}$ and $\underline{w}^\ell_{III}$ denote the highest wage floor such that each entrepreneur is in region II. Using equations 19, 31, and 35,

$$\underline{w}^h_I = \frac{\theta}{1+\theta}\frac{MRPL(z, k^h, n^h_{II})}{P} > \frac{\theta}{1+\theta}\frac{MRPL(z, k(a^\ell), n^\ell_{II})}{P} = \underline{w}^\ell_I,$$

and

$$\underline{w}^h_{III} = \frac{MRPL(z, k^h, n^h_{II})}{P} > \frac{MRPL(z, k(a^\ell), n^\ell_{II})}{P} = \underline{w}^\ell_{III}.$$

□

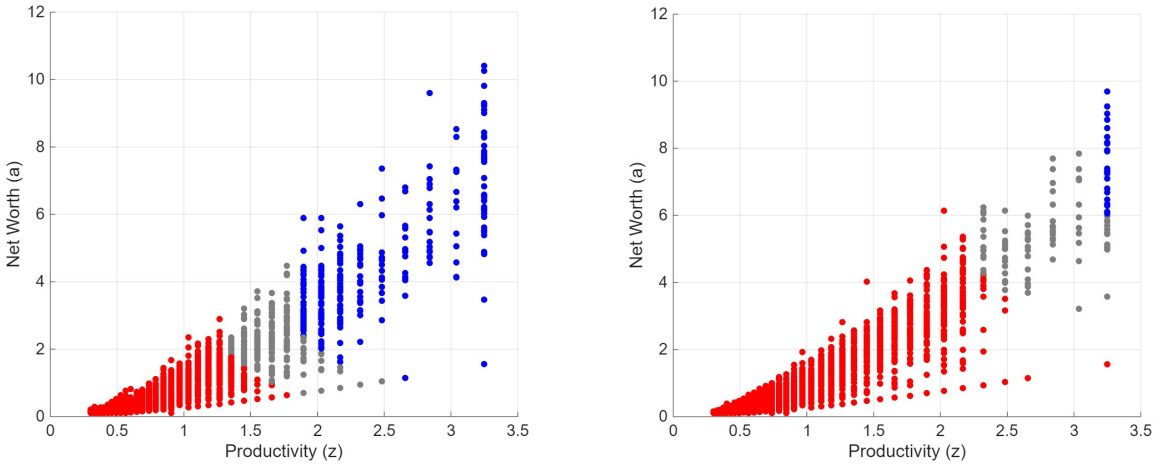
Thus, collateral constraints decrease investment for constrained entrepreneurs and,

due to the impact of lower capital on $MRPL$, decrease labor allocations and the cutoff wages at which an entrepreneur moves across regions. Figure B.1 plots the entrepreneurs' choices of capital and labor as a function of productivity, net worth, and the wage floor.

C Quantitative analysis

C.1 Distribution of entrepreneurs by region

Figure C.1: Distribution of entrepreneurs in each region by productivity and net worth



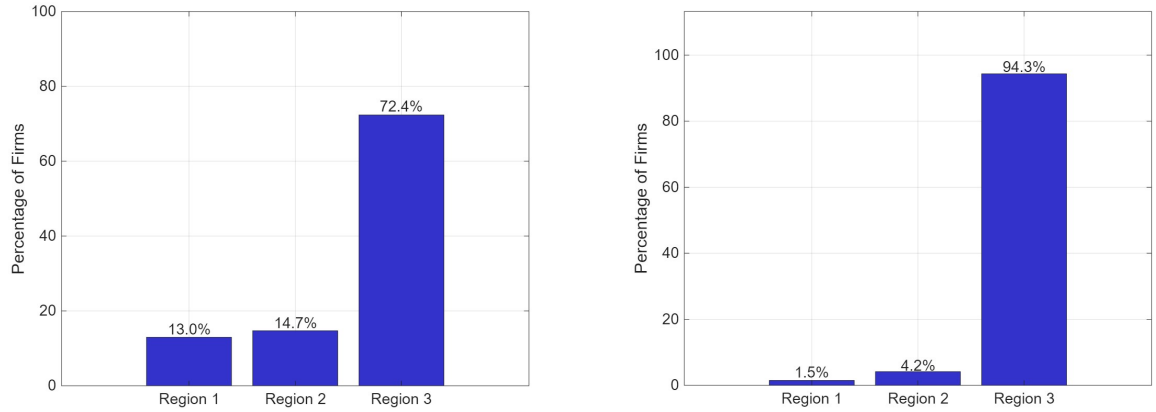
(a) Pre-$\underline{w}\uparrow$ (b) Post-$\underline{w}\uparrow$]

Note: The figure plots the distribution of entrepreneurs across the regions of constraint under $\underline{w}$ by productivity z and net worth a. Red dots indicate region III, grey dots represent region II, and blue dots represent region I. Panel a plots the distribution in the baseline economy, and Panel b plots the distribution after the 20% wage floor increase.

Figure C.1 plots the distribution of entrepreneurs across the three regions of constraint by productivity and net worth. The left panel shows the distribution in the baseline calibration, and the right panel shows the distribution after the wage floor increase. Red dots indicate an entrepreneur in region III, gray dots indicate region II, and blue dots indicate region I. In both calibrations, the bulk of entrepreneurs are in region III, suggesting an inefficient wage floor even before the 20% increase. High-productivity, high-net-worth entrepreneurs are the most likely to be in region I. Looking across the distribution, entrepreneurs with the same productivity but different net worths are sometimes in different regions, with lower-net-worth entrepreneurs tending to be more constrained under a given $\underline{w}$. This reflects the role of collateral constraints in affecting entrepreneur outcomes. Figure C.2 shows the percentage of entrepreneurs in each region under each calibration. In the baseline calibration, 72% of entrepreneurs are already in region III.

Figure C.2: Percent of entrepreneurs in each region

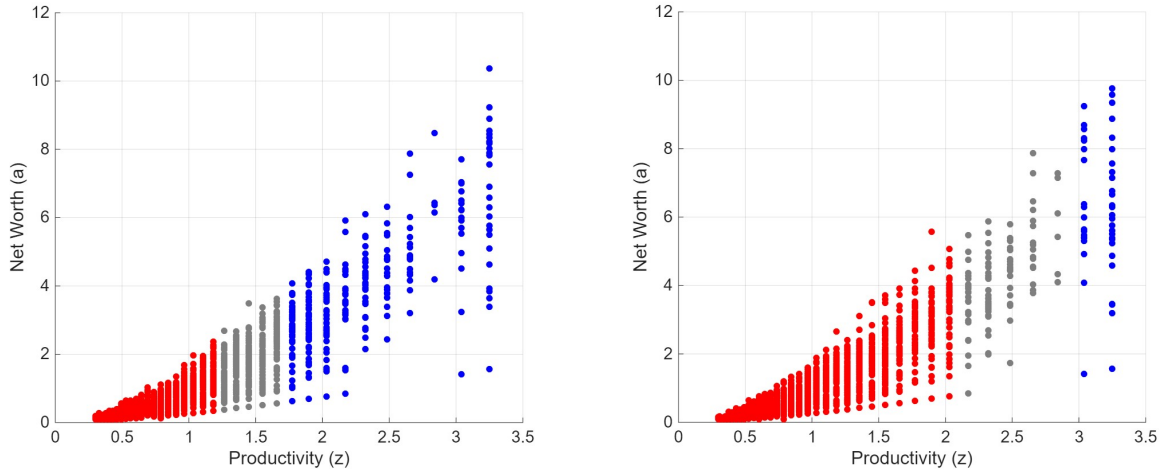
(a) Pre-$\underline{w}\uparrow$ (b) Post-$\underline{w}\uparrow$]



Note: The figure plots the percentage of entrepreneurs in each region of constraint under $\underline{w}$. Panel a plots the distribution in the baseline economy, and Panel b plots the distribution after the 20% wage floor increase.

Following the wage floor increase, this increases to 94%, with only 1.5% of entrepreneurs in region I and 4.4% in region II.

Figure C.3: Distribution of entrepreneurs in each region with no collateral constraint

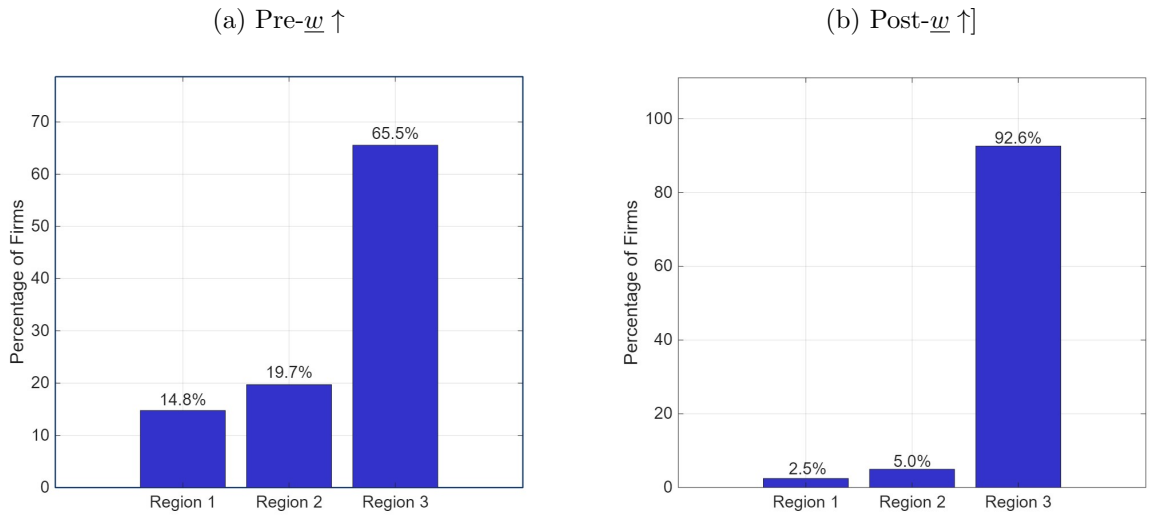


(a) Pre-$\underline{w}\uparrow$ (b) Post-$\underline{w}\uparrow$

Note: The figure plots the distribution of entrepreneurs across the region of constraint under $\underline{w}$ by productivity z and net worth a when collateral constraints are removed from the model, $\chi \to \infty$. Red dots indicate region III, gray dots represent region II, and blue dots represent region I. Panel a plots the distribution in the baseline economy, and Panel b plots the distribution after the 20% wage floor increase.

Role of financial frictions: To get a clearer idea of how financial frictions exacerbate the consequences of wage floor increases, I remove the collateral constraint from the model, $\chi \to \infty$. Figure C.3 plots the distribution of entrepreneurs across the three regions of

Figure C.4: Percent of entrepreneurs in each region with no collateral constraint



Note: The figure plots the percentage of entrepreneurs in each region of constraint under $\underline{w}$ when collateral constraints are removed from the model, $\chi \to \infty$. Panel a plots the distribution in the baseline economy, and Panel b plots the distribution after the 20% wage floor increase.

constraint by productivity and net worth in the environment without collateral constraints, and Figure C.4 reports the percentage of entrepreneurs falling into each region. The left panels show the distribution before the wage floor increase, and the right panels show the distribution after the wage floor increase. Note that, without financial frictions, all entrepreneurs with the same productivity now fall into the same region regardless of net worth. Comparing the results to the distributions with collateral constraints in Figures C.1 and C.2, the majority of entrepreneurs are still in region III before and after the wage floor increase. However, this is less pronounced in the environment with no financial frictions, with 65.5% of entrepreneurs in region III before the wage floor increase (relative to 72.4% with financial frictions) and 92.6% in region III after the increase (relative to 94.3%). Financial frictions make a smaller difference in pushing entrepreneurs into region III in the post-wage floor increase regime because the higher wage floor is so restrictive on its own. However, there can still be a significant role for financial frictions in generating aggregate contractions by pushing entrepreneurs further into region III than they otherwise would be.

C.2 Optimal wage floors

To compute the optimal policy, I follow Berger et al. (2025) and use the consumption-equivalent welfare gain relative to a no-wage-floor economy. I consider optimal wages based upon maximizing worker welfare, so this is the proportionate increase in consumption, $\Lambda^w(\underline{w})$,

that makes workers indifferent between the no-wage floor and wage-floor economies. $\Lambda^w(\underline{w})$ is defined such that

$$u^w\left(\left(1+\Lambda^w(\underline{w})\right)c^w(\underline{w}=0),N(\underline{w}=0)\right)=u^w\left(c^w(\underline{w}),N(\underline{w})\right).$$

Consistent with Berger et al. (2025), I find that potential welfare gains from wage floors are small. Welfare gains top out at at 0.08% when $\underline{w}=0.21$.[23] Removing financial frictions results in similar maximum welfare gains. Still, a slightly higher optimal wage floor, $\underline{w}=0.27$, reflects that fewer entrepreneurs are constrained under any given wage floor in an environment with no financial frictions. In either case, the optimal level is significantly below the initial wage floor in the baseline calibration, suggesting inefficiencies associated with the nationalized bargaining system even before the 20% wage floor increase.

C.3 Solution Algorithm

To solve for the stationary recursive equilibrium, I discretize the productivity process $f(z)$ into $n_z=50$ nodes. The second stage problem can be solved using analytic expressions. The first stage is solved using the endogenous grid method (EGM). I include a sketch of the algorithm below.

1. Guess aggregate quantities and prices: $S=(P,P^Q,Y,Q,X,M,D,\tilde{W},N)$.
2. Given a guess for S, solve the entrepreneur problem for each z
   (a) Determine the entrepreneur's steady state region: $\bar{r}(z)=1$ if $z>\tilde{z}_1$, $\bar{r}(z)=3$ if $z<\tilde{z}_2$ and $\bar{r}(z)=2$ if $z\in[\tilde{z}_2,\tilde{z}_1]$. Where

$$\tilde{z}_2=\left(\frac{\sigma-1}{\sigma}P^Q Q^{\frac{1}{\sigma}}\right)^{\frac{\sigma}{1-\sigma}}(1-\alpha)^{\frac{\sigma+\alpha-\sigma\alpha}{1-\sigma}}\left(\frac{\alpha}{P(i^*+\delta)}\right)^{-\alpha}\left(\frac{N}{\tilde{W}^\theta}\right)^{\frac{1}{\sigma-1}}\underline{w}^{\frac{\alpha+\sigma+\theta-\sigma\alpha}{\sigma-1}}$$

$$\tilde{z}_1=\left(\frac{\sigma-1}{\sigma}P^Q Q^{\frac{1}{\sigma}}\right)^{\frac{\sigma}{1-\sigma}}\left(\frac{(1-\alpha)(1+\theta)}{\theta}\right)^{\frac{\sigma+\alpha-\sigma\alpha}{1-\sigma}}\times\left(\frac{\alpha}{P(i^*+\delta)}\right)^{-\alpha}\left(\frac{N}{\tilde{W}^\theta}\right)^{\frac{1}{\sigma-1}}\underline{w}^{\frac{\alpha+\sigma+\theta-\sigma\alpha}{\sigma-1}}$$

   (b) Given $\bar{r}(z)$ compute steady state capital $\bar{k}(z)$, capital endowments for newborn

[23] Note that the model is not calibrated such that the level of the wage floor is economically meaningful. This does not imply wage floors should optimally be set to 0.21 euro. The optimal level is valuable only in comparison to the starting calibration.

entrepreneurs $\underline{k}(z)$, and lump sum taxes $T(z)$ that balance the government budget constraint $T(z) = \rho p \underline{k}(z)$

i. if $\bar{r}(z) = 1$

$$P(i^* + \delta) = \frac{\sigma - 1}{\sigma} P^Q Q^{\frac{1}{\sigma}} \alpha (z n^{1-\alpha})^{\frac{\sigma-1}{\sigma}} k^{\frac{\alpha\sigma-\sigma-\alpha}{\sigma}}$$

$$n^{ss}(k, z) = \left(\frac{\sigma - 1}{\sigma} \frac{\theta}{\theta + 1} P^Q Q^{\frac{1}{\sigma}} \frac{(1-\alpha)}{\tilde{W}} N^{\frac{1}{\theta}} (z k^{\alpha})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma\theta}{\sigma - \alpha\theta + \theta + \sigma\alpha\theta}}$$

$$k^{ss}(z) = \left(\frac{\sigma - 1}{\sigma} P^Q Q^{\frac{1}{\sigma}} z^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma(1+\theta)}{\alpha+\theta+\sigma-\alpha\sigma}} \times \left(\frac{\alpha}{P(i^* + \delta)} \right)^{\frac{\sigma-\alpha\theta+\theta+\sigma\alpha\theta}{\alpha+\theta+\sigma-\alpha\sigma}} \left(\frac{\theta}{\theta + 1} \frac{(1-\alpha)}{\tilde{W}} N^{\frac{1}{\theta}} \right)^{\frac{(1-\alpha)(\sigma-1)\theta}{\alpha+\theta+\sigma-\alpha\sigma}}$$

ii. if $\bar{r}(z) = 2$

$$P(i^* + \delta) = \frac{\sigma - 1}{\sigma} P^Q Q^{\frac{1}{\sigma}} \alpha (z n^{1-\alpha})^{\frac{\sigma-1}{\sigma}} k^{\frac{\alpha\sigma-\sigma-\alpha}{\sigma}}$$

$$n^{ss} = \left(\frac{w}{\tilde{W}} \right)^{\theta} N$$

$$P(i^* + \delta) = \frac{\sigma - 1}{\sigma} P^Q Q^{\frac{1}{\sigma}} \alpha (z)^{\frac{\sigma-1}{\sigma}} \left(\left(\frac{w}{\tilde{W}} \right)^{\theta} N \right)^{\frac{(\sigma-1)(1-\alpha)}{\sigma}} k^{\frac{\alpha\sigma-\sigma-\alpha}{\sigma}}$$

$$k^{ss}(z) = \left(\frac{\sigma - 1}{\sigma} P^Q Q^{\frac{1}{\sigma}} \frac{\alpha}{p(r^* + \delta)} z^{\frac{\sigma-1}{\sigma}} \left(\left(\frac{w}{\tilde{W}} \right)^{\theta} N \right)^{\frac{(\sigma-1)(1-\alpha)}{\sigma}} \right)^{\frac{\sigma}{\sigma+\alpha-\alpha\sigma}}$$

iii. if $\bar{r}(z) = 3$

$$P(i^* + \delta) = \frac{\sigma - 1}{\sigma} P^Q Q^{\frac{1}{\sigma}} \alpha (z n^{1-\alpha})^{\frac{\sigma-1}{\sigma}} k^{\frac{\alpha\sigma-\sigma-\alpha}{\sigma}}$$

$$n^{ss}(z) = \left(\frac{\sigma - 1}{\sigma} \frac{1-\alpha}{\underline{w}} P^Q Q^{\frac{1}{\sigma}} (z k^{\alpha})^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\alpha\sigma+1-\alpha}}$$

$$P(i^* + \delta) = \left(\frac{\sigma - 1}{\sigma} P^Q Q^{\frac{1}{\sigma}} z^{\frac{\sigma-1}{\sigma}} \right)^{\frac{\sigma}{\sigma\alpha+1-\alpha}} \alpha \left(\frac{1-\alpha}{\underline{w}} \right)^{\frac{(1-\alpha)(\sigma-1)}{\sigma\alpha+1-\alpha}} k^{\frac{-1}{\alpha\sigma+1-\alpha}}$$

$$k^{ss}(z) = \left(\frac{\sigma - 1}{\sigma} P^Q Q^{\frac{1}{\sigma}} z^{\frac{\sigma-1}{\sigma}} \right)^{\sigma} \left(\frac{\alpha}{P(i^* + \delta)} \right)^{\sigma\alpha+1-\alpha} \left(\frac{1-\alpha}{\underline{w}} \right)^{(1-\alpha)(\sigma-1)}$$

(c) Define grids for future net worth $a'_r(z)$ for each $r \in R = 1, 2, 3$ and compute the associated 2nd stage grids for cash-on-hand $\tilde{m}'_r(a', z)$, debt $d'_r(a', z)$, and

collateral multiplier $\eta'_r(a', z)$.

i. Solve for capital

$$k' = \min\{\frac{a'}{1+\theta}, k_1^{ss}(z)\}$$

ii. Solve for debt

$$d'_1(a', z) = P(1+i^*)(k' - a')$$

iii. Solve for labor assuming the entrepreneur is in region II

$$\hat{n}'_{II} = \left(\frac{w}{\tilde{W}}\right)^{\theta} N$$

iv. Use $\hat{n}'_{II}$ and k' to compute the marginal revenue product of labor

$$M\hat{R}PL_{II} = \frac{1-\alpha}{\vartheta} P^Q Q^{\frac{1}{\sigma}} (z(k')^{\alpha})^{\frac{\sigma-1}{\sigma}} (\hat{n}'_{II})^{\frac{\alpha-1-\sigma\alpha}{\sigma}}$$

v. Compute the markdown

$$\hat{\mu}_{II} = \frac{\bar{w}}{M\hat{R}PL_{II}/P}$$

vi. If $\hat{\mu}_{II} < \theta/(1+\theta)$ (region I)

$$n' = \left(\frac{\sigma-1}{\sigma}\frac{\theta}{\theta+1} P^Q Q^{\frac{1}{\sigma}} \frac{(1-\alpha)}{\tilde{W}} N^{\frac{1}{\theta}} (zk^{\alpha})^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma\theta}{\sigma-\alpha\theta+\theta+\sigma\alpha\theta}}$$

$$w = \left(\frac{n}{N}\right)^{\frac{1}{\theta}} \tilde{W}$$

vii. if $\hat{\mu}_{II} > 1$ (region III)

$$n' = \left(\frac{\sigma-1}{\sigma}\frac{1-\alpha}{\underline{w}} p y^{\frac{1}{\sigma}} (zk^{\alpha})^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\alpha\sigma+1-\alpha}}$$

$$w = \underline{w}$$

viii. $\hat{\mu}_{II} \in [\theta/(1+\theta), 1]$ (region II)

$$n' = \left(\frac{w}{\tilde{W}}\right)^{\theta} N$$

$$w = \underline{w}$$

ix. price of intermediate good

$$\varphi = \left(\frac{z(k')^{\alpha}(n')^{1-\alpha}}{(P^Q)^{\sigma} Q}\right)^{-\frac{1}{\sigma}}$$

x. cash-on-hand

$$m(a,z) = \left(\frac{\varphi}{P^Q}\right)^{1-\sigma} Q - wn + (1-\delta)k - d - T(z)$$

xi. collateral constraint multiplier:

$$\eta(a,z) = \frac{1}{1-\theta}\left(\frac{\sigma-1}{\sigma}\varphi\alpha z\left(\frac{k}{n}\right)^{\alpha-1} - P(i^*+\delta)\right)$$

(d) Given $a'(z)$, $\tilde{m}'(a',z)$, $d'(a',z)$, $\eta'(a',z)$ for each $r \in R = 1,2,3$, Solve for $a'_r(m,z)$ and $c^e(m,z)$ using EGM to iterate on the Entrepreneur's first stage Euler equation. Compute the value function $v(m,z)$ through iteration on the converged policy function.

(e) Given $k_0 = \underline{k}(z)$ and $T(z)$ solve the entrepreneur's newborn state: $m_0(z) = \underline{m}(z)$

(f) Map the solutions for the state space (m,z) into (τ,z) by recursive substitution: when $\tau = 0$, $a'(0,z) = a'(m(z),z)$ and $m'(0,z) = m'(a'(0,z),z)$. Then recursively $a'(\tau,z) = a'(m'(\tau-1,z),z)$ and $m'(\tau,z) = m'(a'(\tau,z),z)$.

3. Use the market clearing conditions to construct the system of equations $h(\mathbf{S}) = \epsilon$

$$N - \varsigma(\tilde{W})^{\varsigma} = \epsilon_1$$

$$Q - X - D = \epsilon_2$$

$$X - \bar{X}(P^Q)^{-\psi} = \epsilon_3$$

$$D - \omega Y\left(\frac{P}{P^Q}\right)^{\psi} = \epsilon_4$$

$$M - (1-\omega)YP^{\psi} = \epsilon_5$$

$$Y_t - \left\{\omega^{\frac{1}{\omega}} D_t^{\frac{\psi-1}{\psi}} + (1-\omega)^{\frac{1}{\psi}} M_t^{\frac{\psi-1}{\psi}}\right\}^{\frac{\psi}{\psi-1}} = \epsilon_6$$

$$N_t - \left[\sum_{\tau}\sum_{z} n(\tau,z)^{\frac{\theta+1}{\theta}}\right]^{\frac{\theta}{\theta+1}} = \epsilon_7$$

$$Q - \left(\sum_{\tau}\sum_{z}\left(zk(\tau,z)^{\alpha}n(\tau,z)^{1-\alpha}\right)^{\frac{\sigma-1}{\sigma}}\right)^{\frac{\sigma}{\sigma-1}} = \epsilon_8$$

$$Y_t - \sum_z \rho \underline{k}(z) f(z) - \sum_\tau \sum_z w(\tau, z) n(\tau, z)$$
$$- \sum_\tau \sum_z \left[c^e(\tau, z) + (1 - \rho) k'(\tau, z) - (1 - \delta) k(\tau, z) \right] \phi(\tau, z) = \epsilon_9$$

4. Solve for $h(\mathbf{S}) \approx 0$ using Broyden's method

5. Check for convergence between the guess $\hat{S} = (P, P^Q, Y, Q, X, M, D, \tilde{W}, N)$ used in step 1 and the solution from step 4. If convergence fails, update $\hat{S}$ and return to step 2.